# Water

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# Introduction

**1). Like air, water is one of the few basic materials which is of prime importance for the preservation of life on this earth.**

**2). All are aware of the uses of water for drinking, cooking, cooking, bathing & for farming etc.**

**3). But few know the importance of water as an engineering material.**

**4). As an engineering material water is used for producing steam, in boilers to generate hydro-electric power, furnishing steam for engines, for construction of concrete structures for manufacturing purposes & as a solvent in chemical process.**

**Sources of Water:**

The Main Sources Of Water Are:

1). Rain water

2). River water

3). Spring or well water

4). Sea water

**Types of impurities in water:**

The impurities present in water are classified as:

1). Dissolved impurities: dissolved impurities may organic or inorganic.

Inorganic impurities: the carbonates, bicarbonates, sulphates, chlorides of calcium, magnesium, iron potassium and aluminium.

Organic impurities: Organic water products, amino acids, proteins, etc.

Gases: $O_2$, $CO_2$, Oxides of nitrogen and sulphur, $H_2S$ etc.

2). **Suspended impurities:** It is of two types:

1. Inorganic- sand & clay;
2. Organic- vegetable and animal matter.

3) **Biological Impurities:** Micro-organisms like pathogenic bacteria, fungi, algae, etc.

**DISADVANTAGES OF HARDWATER/ CAUSES OF HARDNESS:**

The following are the disadvantages when hard water is used for various purpose:

(i) DOMESTIC USE:

(a) Washing and Bathing: Hard water does not form lather easily with soap, so soap is wasted

(b) Drinking: Hard water causes bad effects on our digestive system. Sometimes, stone formation takes place in kidneys.

(c) Cooking: The boiling point of water is increased due to the presence of salts. Hence, more fuel and time are required for cooking.

(ii) INDUSTRIAL USE:

(a) Textile Industry: Hard water causes wastage of soap. Precipitates of calcium and magnesium soap adhere to the fabrics and cause problem such difference in color shades, dull shades, patches, etc.

(b) Paper Industry: Calcium and Magnesium salts in water may affect the quality of paper.

(c) Sugar Industry: Water containing sulphates, carbonates, nitrates affects the crystallisation of sugar.

(d) Pharmaceutical Industry: Hard water may form some undesirable products while preparation of pharmaceutical products.

(iii) STEAM GENERATION IN BOILERS: For steam generation, boilers are employed. If hard water is used in boilers, It may lead to the following troubles-

(a) Boiler Corrosion
(b) Scale and Sludge formation.
(c) Priming and Foaming
(d) Caustic embrittlement Pharmaceutical industry

**HARDNESS OF WATER (OR) HARDWATER AND SOFT WATER:**

Hard Water: The water which does not produce lather (or) very little lather with soap is called Hard Water.

Soft Water: Soft water readily produces a lot of lather when mixed with little soap.

The Hardness of water is caused by the presence of dissolved salts such as Bicarbonates, Sulphates, Chlorides and Nitrates of bivalent metal ions like $Ca^{+2}$ & $Mg^{+2}$.

Soap is sodium/ potassium salt of higher fatty acids like stearic, oleic and palmetic acids. When soap is mixed with soft water lather is produced due to stearic acid and sodium stearate.

Na – Stearate + $H_2O$ → NaOH + Stearic Acid

$[C_{17}H_{35}COOH]$

Stearic Acid + Na-Stearate → Formation of lather.

When soap comes in contact with HARD WATER, Sodium stearate will react with dissolved calcium and magnesium salts and produce calcium stearate or magnesium stearate which is white precipitate.

$2Na–\ Stearate + Ca^{+2} \rightarrow Ca–\ Stearate\downarrow + 2Na^{+}$

$[2C_{17}H_{35}COONa] + Ca^{+2} \rightarrow [(C_{17}H_{35}COO)_2Ca]\downarrow + 2Na^{+}$

(Soap) (Soluble) (Insoluble) (Soluble)

$[2C_{17}H_{35}COONa] + Mg^{+2} \rightarrow [(C_{17}H_{35}COO)_2Mg]\downarrow + 2Na^{+}$

(Soluble) (Soluble) (Insoluble) (Soluble)

## TYPES OF HARDNESS

The hardness of water is of two types-

(1) Temporary hardness (or) Carbonate hardness

(2) Permanent hardness (or) Non-Carbonate hardness

(1) Temporary Hardness: Temporary hardness is caused by two dissolved bicarbonate salts $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$. The hardness is called "Temporary Hardness" because it can be removed easily by means of boiling.

$Ca(HCO_3)_2$ + Heating $\rightarrow$ $CaCO_3\downarrow$ + $H_2O$ + $CO_2\uparrow$

$Mg(HCO_3)_2$ + Heating $\rightarrow$ $Mg(OH)_2\downarrow$ + $2CO_2\uparrow$

(2) Permanent Hardness: Permanent hardness of water is due to the dissolved chlorides, sulphates and nitrates of calcium and magnesium. These salts are $CaCl_2$, $CaSO_4$, $Ca(NO_3)_2$, $MgCl_2$, $MgSO_4$, $Mg(NO_3)_2$. These hardness cannot be removed easily by boiling. Hence it is called "Permanent Hardness". Only chemical treatment can remove this hardness.

**Total Hardness Of Water = Temporary Hardness + Permanent Hardness**

**DEGREE OF HARDNESS:**

- The Concentration of hardness as well as non-hardness constituting ions are, usually expressed in the term of "Equivalent amount of $CaCO_3$"
- Since this mode permits the multiplication and division concentration, when required.
- The choice of $CaCO_3$ in particular is due to its molecular weight (m.wt.) is "100" (Equivalent wt. = 50), and
- Moreover, it is insoluble salt that can be precipitated in water treatment.

Therefore, 100 parts by weight of $CaCO_3$ hardness must be equivalent to-

- 162 parts by weight of $Ca(HCO_3)_2$ hardness
- 146 parts by weight of $Mg(HCO_3)_2$ hardness
- 136 parts by weight of $CaSO_4$ hardness
- 111 parts by weight of $CaCl_2$ hardness
- 164 parts by weight of $Ca(NO_3)_2$ hardness
- 120 parts by weight of $MgSO_4$ hardness
- 95 parts by weight of $MgCl_2$ hardness
- 148 parts by weight of $Mg(NO_3)_2$ hardness

| Salt | Molar Mass | Chemical Equivalent | Multiplication Factor |
|---|---|---|---|
| $Ca(HCO_3)_2$ | 162 | 81 | 100/ 162 |
| $Mg(HCO_3)_2$ | 142 | 73 | 100/ 142 |
| $CaSO_4$ | 136 | 68 | 100/ 136 |
| $FeCl_2$ | 127 | 63.5 | 100/ 127 |
| $CaCl_2$ | 111 | 55.5 | 100/ 111 |
| $MgSO_4$ | 120 | 60 | 100/ 120 |
| $MgCl_2$ | 95 | 47.5 | 100/ 95 |
| $CaCO_3$ | 100 | 50 | 100/ 100 |
| $MgCO_3$ | 84 | 42 | 100/ 84 |
| $CO_2$ | 44 | 22 | 100/ 44 |
| $Ca(NO_3)_2$ | 164 | 82 | 100/164 |
| $Mg(NO_3)_2$ | 148 | 74 | 100/ 148 |
| $HCO_3^-$ | 61 | 61 | 100/ 122 |
| $OH^-$ | 17 | 17 | 100/ 34 |
| $CO_3^{2-}$ | 60 | 30 | 100/ 60 |

| Salt | Molar Mass | Chemical Equivalent | Multiplication Factor |
|---|---|---|---|
| $NaAlO_2$ | 82 | 82 | 100/ 164 |
| $Al_2(SO_4)_3$ | 342 | 57 | 100/ 114 |
| $FeSO_4.7H_2O$ | 278 | 139 | 100/ 278 |
| $H^+$ | 1 | 1 | 100/ 2 |
| HCl | 36.5 | 36.5 | 100/ 73 |

The method of calculating degree of hardness is given by the following formula

- **Hardness causing salt in terms of equivalent of $CaCO_3$= (Amount (Mass) of the hardness causing salt x 100)/ Molecular weight of hardness causing salt**

**UNITS OF HARDNESS:**

The 5 different units in which the hardness of water is expressed as given below-

(1) Parts per million (PPM): PPM is the number of parts of $CaCO_3$ equivalent hardness per $10^6$ parts of water.

i.e., 1 PPM = 1 part of $CaCO_3$ equivalent hardness in $10^6$ parts of water.

(2) Milli grams Per Litre (mg/litre): mg/L is the number of milligrams of $CaCO_3$ equivalent hardness present per litre of water.

i.e., 1 mg/L = 1 mg of $CaCO_3$ equivalent hardness in 1L of water.

But 1 L water weighs = 1 kg of water

1 kg = 1000 gms = 1000 x 1000 mg = $10^6$ mg

$\therefore$ 1 mg/L = 1mg of $CaCO_3$ equivalent per $10^6$ mg of water

= 1 part of $CaCO_3$ equivalent per $10^6$ parts of water

$\therefore$ 1 mg/L = 1 ppm

(3) Degree Of Clarke ($^oCl$): It is number of grains (1/7000 lb) of $CaCO_3$ equivalent hardness per gallon (10 lb) of water.

(or) It is defined as the number of parts of $CaCO_3$ equivalent hardness per 70,000 parts of water.

$\therefore$ 1 $^oCl$ = 1 grain of $CaCO_3$ eq. hardness per gallon of water.

(or) 1 $^oCl$ = 1 part of $CaCO_3$ eq. hardness per 70,000 parts of water

$\therefore$ 1 ppm = 0.07 $^oCl$

(4) Degree Of French ($^oFr$): It is the number of parts of $CaCO_3$ equivalent hardness per $10^5$ parts of water.

1 $^oFr$ = 1 part of $CaCO_3$ equivalent hardness per $10^5$ parts of water

$\therefore$ 0.1 $^oFr$ = 1 ppm

Note: The hardness of water can be converted into all the four units by making use of the following interconversion formula-

1 ppm = 1mg/L = 0.07$^oCl$ = 0.1$^oFr$

1 $^oCl$ = 1.43$^oFr$ = 14.3 ppm = 14.3 mg/L

(5) Milliequivalent per Litre (meq/L): It is the number of milliequivalent of hardness present per litre of water.

1 meq/L= 1 meq of $CaCO_3$ per litre of water

= $10^{-3}$ x 50 g of $CaCO_3$ eq. of hardness per litre of water

= 50 mg of $CaCO_3$ eq. of hardness per litre of water

= 50 mg/L of $CaCO_3$ eq.

= 50 ppm

$\therefore$ 1 meq/L= 50 ppm

**Numerical 1:** Convert the total hardness of water samples in ppm and meq/L from the following.

i) 20.23°Cl

ii) 31.3°Fr

**Numerical 2:** A sample of water has hardness 208ppm of $CaCO_3$ equivalents. Find the hardness in terms of

i) mg/L ii) °Fr iii) °Cl iv) meq/L

**Numerical 3:** How many grams of $FeSO_4$ dissolved per litre gives 210.5ppm of hardness? (Fe = 56, S =32, O =16, Ca =40, C = 12)

Soln: Mol. Wt of $CaCO_3$ = 100g and mol. Wt of $FeSO_4$ = 152g

100g of $CaCO_3$ = 152g of $FeSO_4$

100ppm of $CaCO_3$ = 152ppm of $FeSO_4$

Hence, 210.5ppm of hardness = 152 x 210.5/100 ppm of $FeSO_4$

= 319.96 mg/L

=0.320g/L of $FeSO_4$

**PROBLEM:**

(1) A sample of water gives an analysis 13.6 mg/L of $CaSO_4$ , 7.3 mg/L of $Mg(HCO_3)_2$ . Calculate the total hardness and permanent hardness.

Sol:

| Salt | Quantity (mg/L) | M. Wt. | Multiplication Factor | Eq. of $CaCO_3$ | Hardness |
|---|---|---|---|---|---|
| $CaSO_4$ | 13.6 | 136 | 100/ 136 | 13.6 x (100/ 136)= 10 | P |
| $Mg(HCO_3)_2$ | 7.3 | 146 | 100/ 146 | 7.3 x (100/ 146)= 5 | T |

Total hardness= Temporary hardness + Permanent Hardness = 5 + 10 = 15 mg/L = 15 ppm

Permanent hardness = 10 ppm

Temporary hardness= 5 ppm

**Numerical 4:** Calculate Temporary hardness and permanent hardness of water sample from the following data: $Mg(HCO_3)_2$ = 16.8mg/L, $MgCl_2$ = 19mg/L, $MgSO_4$ = 24mg/L, $Mg(NO_3)_2$ = 29.6mg/L, $CaCO_3$ = 4mg/L, $MgCO_3$ = 10mg/L.

Solution:

| Constituent (mg/L) | Multiplication Factor | $CaCO_3$ Eq. | Hardness |
|---|---|---|---|
| 1. $Mg(HCO_3)_2$ = 16.8 | 50/73 | 11.5 | T |
| 2. $MgCl_2$ = 19 | 50/47.5 | 20 | P |
| 3. $MgSO_4$ = 24 | 50/60 | 20 | P |
| 4. $Mg(NO_3)_2$= 29.6 | 50/74 | 20 | P |
| 5. $CaCO_3$ = 4 | 1 | 4 | T |
| 6. $MgCO_3$ = 10 | 50/42 | 11.9 | T |

Temporary Hardness = hardness due to $Mg(HCO_3)_2$ +$CaCO_3$ + $MgCO_3$

= 11.5 + 4 + 11.9 = 27.4 ppm or 27.4 mg/L

Permanent Hardness = hardness due to $MgCl_2$ + $MgSO_4$ + $Mg(NO_3)_2$

= 20 + 20+ 20 = 60mg/L = 60ppm

**Numerical 5:** A water sample analysis has been found to contain following impurities in ppm: $Mg(HCO_3)_2$ =14.6, $Mg(NO_3)_2$ = 29.6, $Ca(HCO_3)_2$= 8.1, $MgCl_2$ = 19, $MgSO_4$ = 24. calculate the temporary & permanent hardness of water.

**Numerical 6:** How many grams of $CaCl_2$ dissolved per litre gives 150ppm hardness?

**Numerical 7:** Classify the following impurities into temporary permanent and non-hardness causing impurities.

$Ca(HCO_3)_2$, $MgSO_4$, $CaCl_2$, $CO_2$, HCl, $Mg(HCO_3)_2$

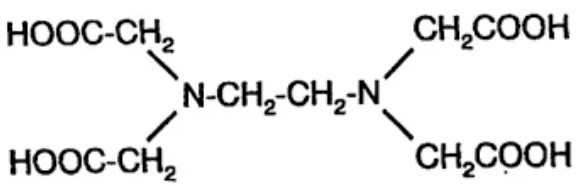
**DETERMINATION OF HARDNESS OF WATER BY EDTA METHOD:-**

1. This is a Complexometric titration method where Ethylenediamine tetraacetic acid (EDTA) is used.
2. EDTA forms complexes with different metal ions at different pH.
3. Calcium & Magnesium ions form complexes with EDTA at pH 9- 10.

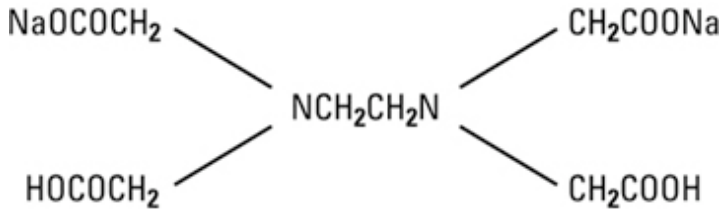
To maintain the pH 9- 10 $NH_4Cl$,$NH_4OH$ buffer solution is used.

4. The disodium salt of EDTA is used for complexation.

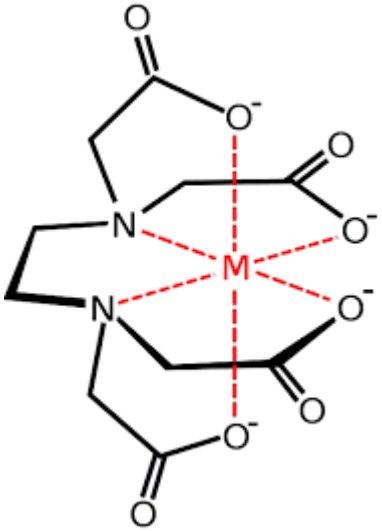
5. An alcoholic solution of Eriochrome Black-T (EBT) is used as an indicator.




EDTA




EDTA disodium salt




Metal-EDTA complex

## BASIC PRINCIPLE:

When hard water comes in contact with EDTA, at pH 9- 10, the $Ca^{+2}$ & $Mg^{+2}$ forms stable, colourless complex with EDTA.

$Ca^{+2}$ + EDTA + pH 9-10 → Ca-EDTA

$Mg^{+2}$ + EDTA + pH 9-10 → Mg-EDTA

**Working:-**

To the hard water sample, the alcoholic blue coloured EBT indicator is added along with the $NH_4Cl$ + $NH_4OH$ buffer (pH 9-10) solution.

EBT forms an unstable wine-red complex with $Ca^{+2}$ & $Mg^{+2}$.

$(Ca^{+2} + Mg^{+2})$ + EBT + Buffer (pH 9-10) → (Ca-EBT + Mg-EBT) complex
(Hard water) (Indicator) (Unstable **wine-red** complex)

This wine-red coloured [Ca-EBT & Mg-EBT] complex is then titrated against EDTA; when EDTA replaces EBT from Ca-EBT & Mg-EBT complex and forms stable colourless [Ca-EDTA] & [Mg-EDTA] complex releasing the blue coloured indicator EBT into $H_2O$.

[Ca-EBT + Mg-EBT] complex + EDTA → EBT + [Ca-EDTA + Mg-EDTA] complex
(Unstable **wine-red** complex) (**Blue**) (Stable colourless complex)

Hence the colour change at the end point is wine-red to blue colour.

**Experimental Procedure:**

**A) Preparations of solutions**

**i) Preparation of Standard Hard water (SHW):**

Dissolve 1g of $CaCO_3$ in minimum amount of dil. HCl till all solid dissolves and evaporated to dryness. Dissolve this residue in 1L distilled water. Each mL of this solution contains 1mg of $CaCO_3$ equivalent hardness.

- Molarity of standard hard water solution =

$$\frac{\text{wt. of } CaCO_3}{\text{M.wt. of } CaCO_3} = \frac{1}{100} = 0.01M$$

1g of $CaCO_3$ in 1L of water= 1000mg of $CaCO_3$ in 1L = 1000ppm soln

1mL of SHW ($CaCO_3$) soln = 1mg of $CaCO_3$

100mL of SHW = _100__mg of $CaCO_3$

1ppm = 1mg/L

**ii) Preparation of EDTA solution**: Dissolve 3.72g of pure EDTA + 0.1 mg $MgCl_2$ in 1L of distilled water.

**iii) Preparation of indicator:** dissolve 0.5g of EBT indicator in 100ml of alcohol.

**iv) Preparation of Buffer soln**: Add 67.5g of $NH_4Cl$ to 570mL of ammonia soln and then dilute with distilled water to 1L.

B) **Procedure: it has three steps:**

**Step 1: Standardization of Hard water:** Pipette out 50mL of SHW in a conical flask. Add 10mL of buffer soln and 4 to 5 drops of EBT indicator. Titrate with EDTA soln till wine red colour changes to blue. Let the volume used be V1 mL.

50mL of SHW = V1 mL of EDTA

1ml of EDTA = ___mg of $CaCO_3$

**Step 2: Titration of unknown hard (bore well) water**: Titrate 50mL of sample water just same as given instep 1. Let the volume used be V2 mL.

50mL of sample HW = V2 mL of EDTA

Total Hardness

**Step 3: Titration of permanent hardness**: Take 50mL of water sample in a beaker and boil it till volume reduces to half. (where all bicarbonates are decomposed to insoluble $CaCO_3$ and $Mg(OH)_2$.) Filter the solution and titrate the filtrate as given in step 1. Let the volume used be V3 mL.

50ml of boiled sample HW = V3 mL of EDTA

Permanent Hardness

**Calculations**:

50ml of SHW = V1 mL of EDTA soln

Since 1 mL SHW contains 1mg of $CaCO_3$,

50mg of $CaCO_3$ = V1 mL of ETDA soln.

Hence, **1mL of EDTA = 50/V1 mg of $CaCO_3$ eq.**

**A) Total Hardness**

50mL of sample hard water = V2 mL of EDTA soln

= V2 x 50/V1 mg of $CaCO_3$ eq.

Hence 1000mL of sample hard water = 1000 V2x 50/V1 x 1/50 mg of $CaCO_3$ eq.

there fore, **Total Hardness = V2/V1 x 1000 ppm**

**B) Permanent Hardness:**

Now, 50ml of boiled sample water = V3 mL of EDTA

= V3 x 50/V1 mg of $CaCO_3$

Hence 1000mL of Sample hard water = V3/V1 x 1000 mg of $CaCO_3$

Therefore, Permanent hardness = V3/V1 x 1000 ppm

**C) Temporary hardness = Total hardness – Permanent hardness**

**Problem:**

1 gm of $CaCO_3$ was dissolved in HCl and the solution was made upto 1 Lit with distilled water. 50 ml of the above solution required 30 ml of EDTA solution for titration. 50 ml of hard water sample required 40 ml of the same solution of EDTA for titration. 50 ml of the hard water after boiling, filtering, etc. required 30 ml of the same EDTA solution for titration. Calculate the temporary hardness of the water.

**Numerical 1:** 50mL of standard hard water requires 38mL of EDTA solution. 100 mL of water sample consumes 21mL of EDTA soln. 100mL of boiled water sample consumes 10mL of EDTA soln. Find all types of hardness.

Solution:

50 mL of SHW = _50_mg of $CaCO_3$

50mL of SHW = 38 ml of EDTA soln.

There fore, 38mL of EDTA = 50mg of $CaCO_3$

1mL of EDTA = _50/38__mg of $CaCO_3$

**1mL of EDTA = 1.316mg of $CaCO_3$**

**Total Hardness:**

100mL of sample HW =21 mL of EDTA solution

= 21 x 1.316 mg of $CaCO_3$

= 27.636 mg of $CaCO_3$

1000mL of sample HW = 27.636 x 1000/100

= 276.36mg of $CaCO_3$/L = 276.36ppm

There fore **Total hardness = 276.36ppm**

**Permanent hardness:**

100mL of Boiled sample HW = 10mL of EDTA

= 13.16 mg of $CaCO_3$

1000mL of boiled sample HW = 13.16 x 1000/100 = 131.6mg of $CaCO_3$ / L

**Permanent hardness = 131.6ppm**

**Temporary Hardness = Total hardness – permanent hardness**

**= 276.36 – 131.6**

**= 144.76ppm**

**Numerical 2:** 50 mL of Standard hard water required 21mL of EDTA solution. 50mL of sample Water requires 21mL of EDTA solution. 50mL of boiled sample water requires 7mL of EDTA soln. calculate all the types of hardness.

Soln:

50mL of SHW = 50mg of $CaCO_3$

50mL of SHW = 21mL of EDTA

There fore 21 mL of EDTA soln = 50mg of $CaCO_3$

Hence **1mL of EDTA soln = 50/21 = 2.381mg of $CaCO_3$ eq.**

**Total Hardness**

50mL of sample HW = 21mL of EDTA soln

= 21 x 2.381 = 50.001 mg of $CaCO_3$ eq.

1000mL of sample HW = 50.001 x 1000/50 = 50.001 x 20

= 1000.02mg $CaCO_3$/L

**Hence Total hardness = 1000.02ppm**

**Permanent Hardness**

50mL of boiled sample HW = 7mL of EDTA soln

= 7 x 2.381 = 16.667mg of $CaCO_3$/L

1000mL of boiled sample HW = 16.667 x 1000/50 = 16.667 x 20 =333.34mg of $CaCO_3$

**Hence permanent hardness = 333.34ppm**

**Temporary Hardness = total hardness – permanent hardness = 1000.02 -333.34 = 666.68 ppm**

**Numerical 3:** 0.38g of $CaCO_3$ was dissolved in HCl and the solution made up to 1Litre with distilled water. 100 mL of the above soln required 25mL of EDTA soln. 100 mL of hard water sample required 37mL of EDTA soln. 100mL of boiled hard water sample required 10mL of EDTA soln. Calculate each type of hardness.

Soln

~~1ml of SHW = 1mg of $CACO_3$~~

SHW = 0.38g of $CaCO_3$/ 1000mL = 380mg of $CaCO_3$ / 1000mL

**1mL of SHW = 0.38mg of $CaCO_3$……using 1st statement of numerical**

100mL of SHW = 25mL of EDTA

100mL of SHW = 38 mg of $CaCO_3$

There fore 25mL of EDTA = 38mg of $CaCO_3$

Hence **1mL of EDTA = 38/25 =1.52mg of $CaCO_3$ ……using 2nd statement of numerical**

**Total Hardness**

100mL of sample HW = 37mL of EDTA

= 37 x 1.52 = 56.24 mg of $CaCO_3$

1000ml of sample HW = 56.24 x 1000/100 = 562.4 mg of $CaCO_3$ /L

**Hence Total hardness = 562.4ppm**

**Permanent hardness**

100mL of Boiled sample HW = 10 ml of EDTA

= 10 x 1.52 = 15.2 mg of $CaCO_3$

1000mL of Boiled sample HW = 15.2 x 1000/100 = 152 mg of $CaCO_3$

**Hence permanent hardness = 152ppm**

**Temporary Hardness = total Hardness – permanent hardness = 562.4-152 = 410.4ppm**

**Numerical 4:** 1.2g of $CaCO_3$ was dissolved in HCl and the solution made up to 1Litre with distilled water. 50 mL of the above soln required 30 of EDTA soln. 100 mL of hard water sample required 27mL of EDTA soln. 100mL of boiled hard water sample required 11mL of EDTA soln. Calculate each type of hardness.

Soln:

**Numerical 5:** 0.55g of $CaCO_3$ was dissolved in HCl and the solution made up to 1Litre with distilled water. 20 mL of the above soln required 10 of EDTA soln. 50 mL of hard water sample required 17mL of EDTA soln. 10mL of boiled hard water sample required 1.1mL of EDTA soln. Calculate each type of hardness.

Ans:

**Numerical 6:** 50mL sample of water require 10.2mL of M/20 disodium EDTA for titration. After boiling and filtration the same volume required 6.5mL of EDTA. Calculate all the types of hardness in it.

**Numerical 6:** 50mL sample of water require 10.2mL of M/20 disodium EDTA for titration. After boiling and filtration the same volume required 6.5mL of EDTA. Calculate all the types of hardness in it.

Soln:

1mL of 0.05M EDTA = ___mg of $CaCO_3$

M/20 EDTA = 1/20M EDTA = 0.05M EDTA

Std relationships says that,

**1000mL of 1M EDTA = 100g of $CaCO_3$……due to Avogadro's numbers of molecules**

1mL of 1M EDTA = 100mg of $CaCO_3$

**1mL of 0.05M EDTA = 100 x 0.05 = 5mg of $CaCO_3$**

**Total Hardness:**

**50 mL of sample HW = 10.2mL of 0.05M EDTA soln**

**= 10.2 x 5 =51mg of $CaCO_3$ eq.**

**1000mL of sample HW = 51 x 1000/50 = 51 x 20 = 1020mg of $CaCO_3$**

**Hence Total Hardness = 1020ppm**

**Permanent Hardness:**

**50mL of Boiled sample HW = 6.5mL of 0.05M EDTA**

**= 6.5 x 5 = 32.5mg of $CaCO_3$ eq.**

**1000mL of boiled Sample HW = 32.5 x 20 = 650mg of $CaCO_3$ eq**

**Hence Permanent Hardness = 650ppm**

**So, temporary hardness = 1020-650 = 370ppm**

**Numerical 7:** 50mL sample of water require 15.3mL of M/15 disodium EDTA for titration. After boiling and filtration the same volume required 9.5mL of EDTA. Calculate all the types of hardness in it.

Solution

**BOILER TROUBLES:**

Water finds a great use in various industries for generation of steam in boilers. When water is continuously evaporated to generate steam, the concentration of the dissolved salts increase progressively causing bad effects for steam boilers. The following are the boiler troubles that arise:

(1) Priming and foaming
(2) Caustic embrittlement
(3) Boiler corrosion
(4) Scale and sludge formation

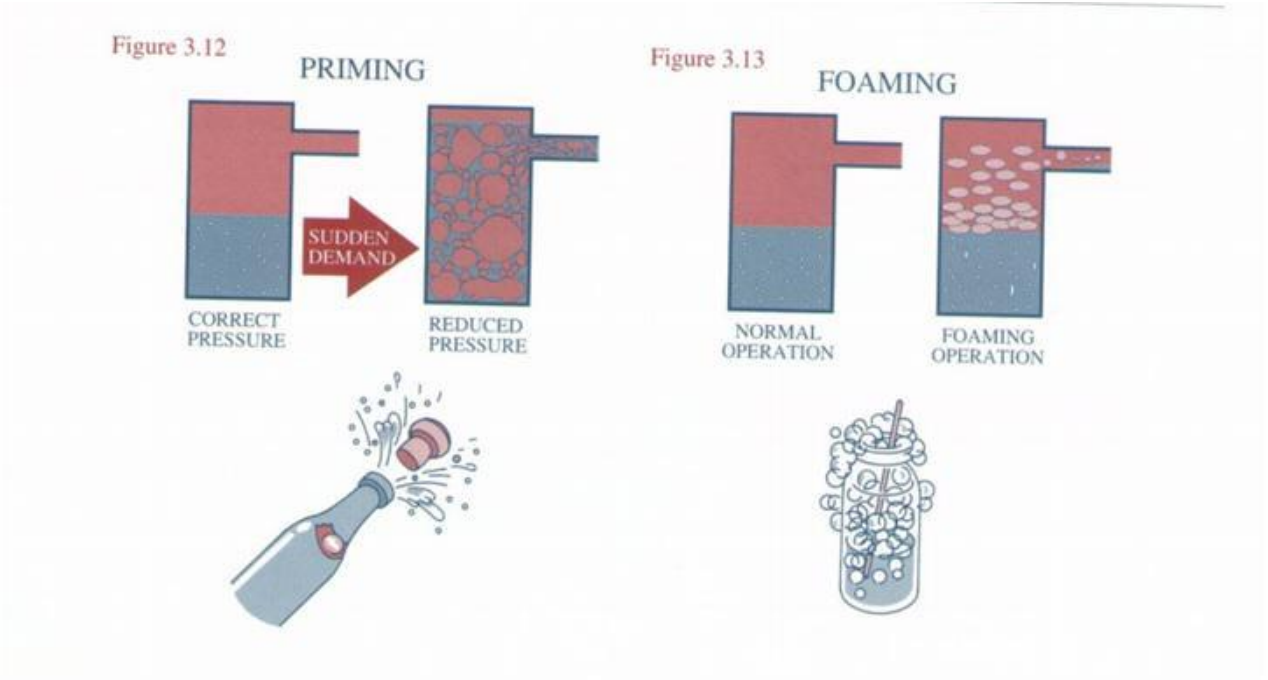
## PRIMING & FOAMING:

Priming: When a boiler is steaming rapidly, some particles of the liquid water are carried along with the steam. This process of "WET STEAM" formation is called 'PRIMING'. Priming is caused by:

(1) Presence of large amount of dissolved solids
(2) High steam velocities
(3) Sudden boiling
(4) Improper boiler design
(5) Sudden increase in steam production rate

It can be avoided by: (Preventions)

(1) Maintaining low water level
(2) Using softened water
(3) Fitting mechanical steam purifiers
(4) Using a well-designed boiler
(5) Avoiding rapid change in steam rate
(6) Blow-down of the boiler



## FOAMING:

Foaming is phenomenon of formation of foam or bubbles on the surface of water inside the boiler with the result that the foam may pass along with the steam.

Causes: The presence of large quantity of suspended impurities and oils lowers the surface tension producing foam.

Preventions: Foaming can be avoided by-

(1) Adding anti foaming chemicals like castor oil
(2) Removing oil from boiler water by adding compounds like "$NaAlO_2$ "
(3) Blow down of the boiler can prevent the foaming

## Disadvantages Of Priming & Foaming:

Priming & Foaming may cause the following boiler troubles:-

(1) The actual height of the water in boiler is not judged

(2) Wastage of heat with the result that it becomes difficult to keep up steam pressure and efficiency of the boiler is lowered

# SLUDGE AND SCALE FORMATION

- In **boilers**, conc. of salts increases due to continuous evaporation of water.
- When the conc. reaches saturation point it is thrown out of the water in form of ppt, which gets settled on the inner walls of the boiler.
- If the ppt takes place in form of loose and soft ppt it is called as **sludge**. It is formed at comparatively colder regions of boiler or at bends. Eg. **$Mg(OH)_2$**
- If the ppt sticks firmly to the inner walls of the boiler it called as **scales**. Sometimes i is even difficult to remove with hammer or chisel. Eg. **$MgSiO_3$**
- **DISADVANTAGES OF SLUDGE AND SCALE FORMATION**

1. Wastage of fuel
2. Lowering of boiler safety
3. Decrease in efficiency
4. Danger of explosion

# SLUDGE AND SCALE FORMATION

**SLUDGE**



**SCALE**



# CAUSTIC EMBRITTLEMENT

- In a boiler, some qty. of $Na_2CO_3$ added to soften the water. But since high pressure and high temperature in the boiler converts $Na_2CO_3$ to NaOH.

  **REACTION: Na2CO3 + H2O → 2NaOH + CO2**

- Because of NaOH, boiler water becomes alkaline. This very dilute alkaline water flows through minute hair cracks.
- When water evaporates, this dilute alkali becomes concentrated.
- This concentrated alkali reacts with **iron** to form **sodium ferroate.**

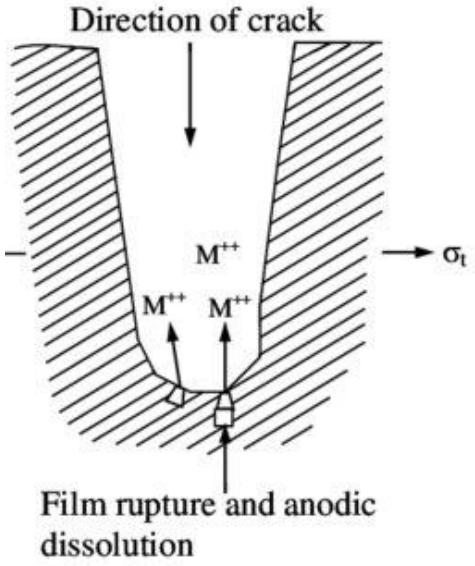
  **REACTION: $2NaOH + Fe \rightarrow Na_2FeO_2 + H_2$**

- This sodium ferroate gets deposited in crevice and cracks.
- This reaction is reversible i.e. where at high temperature sodium ferroate decomposes to form NaOH and $Fe_3O_4$. So reaction continues making use of iron metal of boiler.



## CAUSTIC EMBRITTLEMENT:

Caustic embrittlement is a term used for the appearance of cracks inside the boiler particularly at those places which are under stress such as rivetted joints due to the high concentration of alkali leading to the failure of the boiler. The cracks have appearance of brittle fracture. Hence, the failure is called “Caustic Embrittlement”



Reasons for the formation of Caustic Embrittlement:

The boiler feed water containing carbonates and bicarbonates of alkali metals, sodium hydroxide (NaOH) and a small quantity of silica or sodium silicate; it is purified by Lime-Soda Process.

During the softening process by lime soda process, free $Na_2CO_3$ is usually present in small portion in the soft water which decomposes to give NaOH and $CO_2$ at high pressure of the boilers.

**$Na_2CO_3 + H_2O \rightarrow 2NaOH + CO_2$**

The precipitation of NaOH makes the boiler water **"Caustic"**

The NaOH containing water flows into the small pits and minute haircracks present on the inner walls of the boiler.

As the water evaporates, the concentration of caustic soda (NaOH) increases progressively, creating a **"Concentration Cell"**

Thus, dissolving the iron of the boiler as Sodium Ferrate.

**$Fe + 2NaOH \rightarrow Na_2FeO_2 + H_2$**

The cracking of the boiler occurs particularly at stressed parts like bends, joints, rivets, etc. causing the failure of the boiler.

The **iron** at plane surfaces **surrounded by dilute NaOH** becomes **Cathodic**; while the **Iron** at bends, rivets and joints is surrounded by **highly concentrated NaOH** becomes **Anodic** which consequently decayed or corroded

(Anodic site) Conc NaoH || dil NaoH (Cathodic site)

Iron at Joints &bends  Iron at Plane surface

<u>Prevention Of Caustic Embrittlement:</u>

(1) By adding $Na_2SO_4$ , tannin, etc. to the boiler water which blocks hair cracks. There by preventing infiltration of caustic soda solution

(2) By using sodium phosphate as the softening agent instead of sodium carbonate

## DISADVANTAGES OF CAUSTIC EMBRITLLEMENT:

The cracking or weakening of the boiler metal causes failure of the boiler

(3) BOILER CORROSION: The decay of boiler material by chemical/ electro- chemical attack by its environment is called “Boiler Corrosion”.
Reasons for boiler corrosion are: (a) Dissolved oxygen (b) Dissolved $CO_2$ (c) Acids from dissolved salts.

(a) Dissolved Oxygen: Water usually contains about 8 mg/L of dissolved oxygen at room temperature. Dissolved $O_2$ at high temperature attacks boiler material.

**$2Fe + 2H_2O + O_2 \rightarrow 2Fe(OH)_2\downarrow$ (ferrous hydroxide)**

**$4Fe(OH)_2\downarrow + O_2 \rightarrow 2[Fe_2O_3.2H_2O]\downarrow$ (ferric oxide)**

Removal of dissolved $O_2$ :

By adding calculated quantity of sodium sulphate (or) hydrazine (or) sodium sulphide

$2Na_2SO_3 + O_2 \rightarrow 2Na_2SO_4$

Sodium Sulphite

$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$

Hydrazine

$Na_2S + 2O_2 \rightarrow Na_2SO_4$

(b) Dissolved $CO_2$:

Dissolved $CO_2$ has slow corrosive effect on the materials of boiler plate. Source of $CO_2$ into water is the boiler feed water which contains bicarbonates. Under the high temperature and pressure, maintained in the boiler the bicarbonates decompose to produce $CO_2$.

**$Ca(HCO_3)_2 \rightarrow CaCO_3 + CO_2 \uparrow + H_2O$**

**$Mg(HCO_3)_2 \rightarrow Mg(OH)_2 + 2CO_2 \uparrow$**

The disadvantage of the $CO_2$ is slow corrosive effect on boiler plates by producing **carbonic acid**

**$CO_2 + H_2O \rightarrow H_2CO_3$**

Removal of $CO_2$:

By the addition of calculated quantity of ammonia.

$2NH_4OH + CO_2 \rightarrow (NH_4)_2CO_3 + H_2O$

(c) Acids from dissolved salts:

Water containing dissolved Mg-salts liberate acids on hydrolysis

**$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2\downarrow + 2HCl$**

Disadvantages of the acid production is that the acids react with Iron of the boiler plate in a chain reaction to produce decay of the metal.

**$Fe + 2HCl \rightarrow FeCl_2 + H_2$**

**$FeCl_2 + H_2O \rightarrow Fe(OH)_2 + 2HCl$**

**$Fe(OH)_2 + H_2O + 1/2O_2 \rightarrow Fe_2O_3 .3H_2O$ (ferric oxide)**

Consequently even a small amount of $MgCl_2$ can cause corrosion to a large extent.

Preventions:

(1) Softening of boiler water to remove $MgCl_2$ from the water
(2) Addition of corrosion inhibitors like sodium silicates, sodium phosphate & sodium chromate
(3) By frequent **blow-down operation** i.e., removal of water, concentrated with dissolved salts and feeding the boiler with fresh soft water.
(4) Sludges and Scales formation: In boiler, water evaporates continuously and the concentration reaches saturation point, they form precipitates (scale or sludge) on the inner wall of the boiler

**SOFTENING OF WATER:**

The removal of hardness causing salts from water is called “Softening of water”.

The 3 Industrial methods employed for softening of water are:

(1) Lime-Soda Process

(2) Zeolite (or) Permutite Process

(3) Ion-Exchange (or) Demineralization process

## (1) LIME-SODA PROCESS:

This process is based on converting the soluble calcium and magnesium salts into Insoluble calcium carbonate and magnesium hydroxide precipitates by addition of calculated amount of lime ($Ca(OH)_2$) and Soda ($Na_2CO_3$). The precipitate are removed by filtration. Any free dissolved $CO_2$ and acids are also removed by this process. The various chemical reactions involved in this process are:

(a) For Calcium and Magnesium bicarbonates, only lime is required

(i) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$

(ii) $Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + Mg(OH)_2\downarrow + 2H_2O$

(b) For $MgSO_4$ & $MgCl_2$ , both lime & soda are required

(iii) $MgSO_4 + Na_2CO_3 + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + CaCO_3\downarrow + Na_2SO_4$

(iv) $MgCl_2 + Na_2CO_3 + Ca(OH)_2 \rightarrow Mg(OH_2)\downarrow + CaCO_3 + 2NaCl$

(c) For $CaSO_4$ & $CaCl_2$, only Soda is required

(v) $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$

(vi) $CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2NaCl$

(d) Other Reactions: Free acids, CO2 , H2S dissolved iron and aluminium salts etc are also removed in this process

$2HCl + Na_2CO_3 \rightarrow 2NaCl + H_2O + CO_2\uparrow$

$H_2SO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + H_2O + CO_2\uparrow$

$CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$

$H_2S + Ca(OH)_2 \rightarrow CaS\downarrow + 2H_2O$

$FeSO_4 + Ca(OH)_2 \rightarrow Fe(OH)_2\downarrow + CaSO_4$

$Al_2(SO_4)_3 + 3Ca(OH)_2 \rightarrow 2Al(OH)_3\downarrow + 3CaSO_4 + H_2O$

| Constituent | Reaction | Requirement |
|---|---|---|
| $Ca^{+2}$ (Permanent) | (i) $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$<br>(ii) $CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2NaCl$ | S |
| $Mg^{+2}$ (Permanent) | (iii) $MgSO_4 + Na_2CO_3 + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + CaCO_3\downarrow + Na_2SO_4$<br>(iv) $MgCl_2 + Na_2CO_3 + Ca(OH)_2 \rightarrow Mg(OH_2)\downarrow + CaCO_3 + 2NaCl$ | L + S |
| $HCO3^-$ | (v) $2NaHCO_3 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O + Na_2CO_3$ | L - S |
| $Ca(HCO_3)_2$ (Temporary) | (vi) $Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$ | L |
| $Mg(HCO_3)_2$ (Temporary) | (vii) $Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + Mg(OH)_2\downarrow + 2H_2O$ | 2L |
| $CO_2$ | (viii) $CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$ | L |
| $H^+$ | (ix) $2HCl + Ca(OH)_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaCl + H_2O$ | L + S |
| Coagulants $FeSO_4$ | (x) $FeSO_4 + Ca(OH)_2 \rightarrow Fe(OH)_2 + CaSO_4$<br>$2Fe(OH)_2 + H_2O + O_2 \rightarrow 2Fe(OH)_3$<br>$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$ | L + S |
| $Al_2(SO_4)_3$ | (xi) $Al_2(SO_4)_3 + 3Ca(OH)_2 \rightarrow 2Al(OH)_3\downarrow + 3CaSO_4 + H_2O$<br>$3CaSO_4 + 3Na_2CO_3 \rightarrow 3CaCO_3 + 3Na_2SO_4$ | L + S |

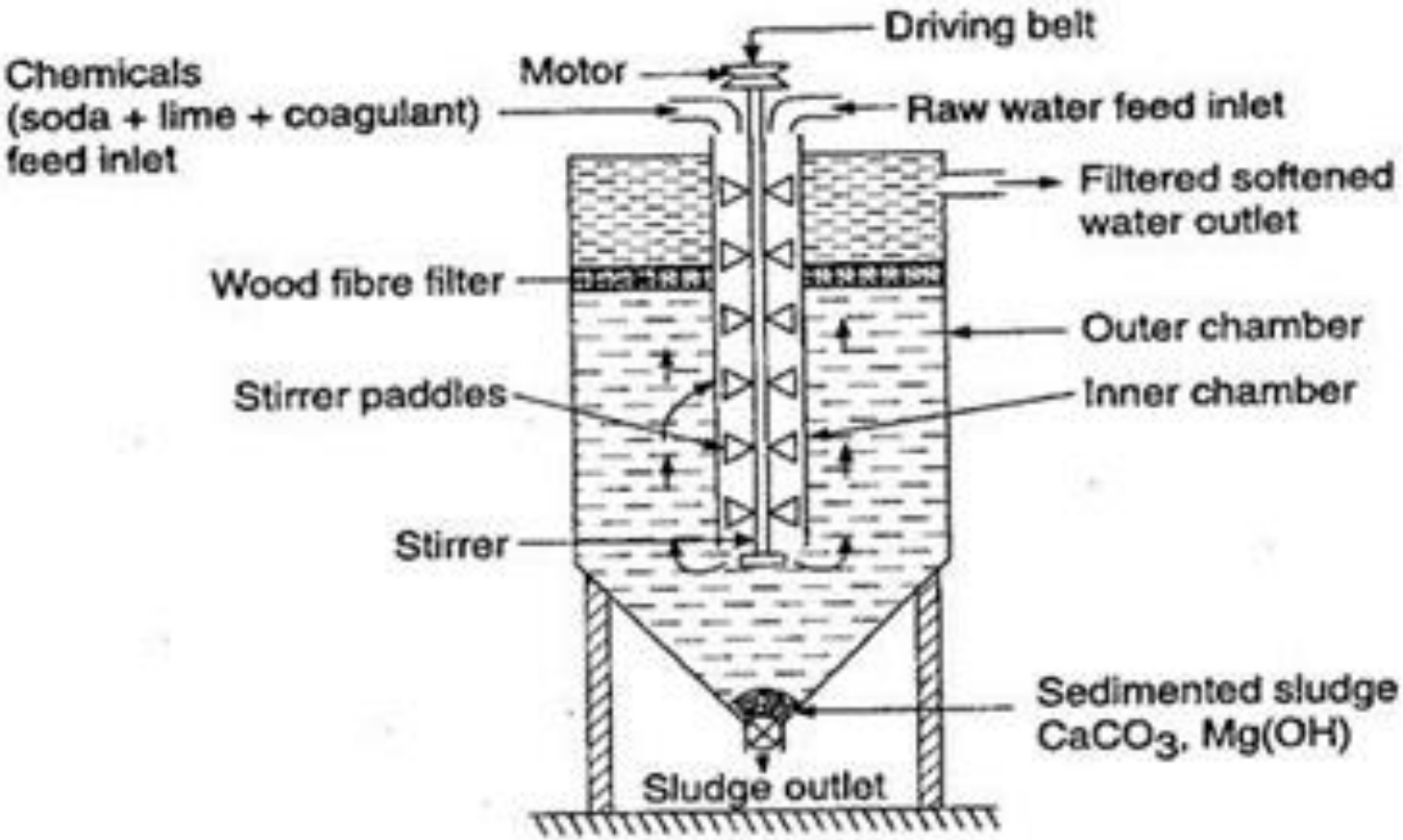
## Calculation:

100 parts by mass of $CaCO_3$ are equivalent to 74 parts of $Ca(OH)_2$ and 106 parts of $Na_2CO_3$.

(a) Amount of lime required for softening = 74/100 (Temp $Ca^{2+}$ + 2 x Temp. $Mg^{2+}$ + Perm. ($Mg^{+2}$ + $Fe^{+2}$ + $Al^{3+}$) + $CO_2$ + $H^+$ ($HCl/H_2SO_4$ ) + $HCO_3$ - all in terms of $CaCO_3$ equivalent)

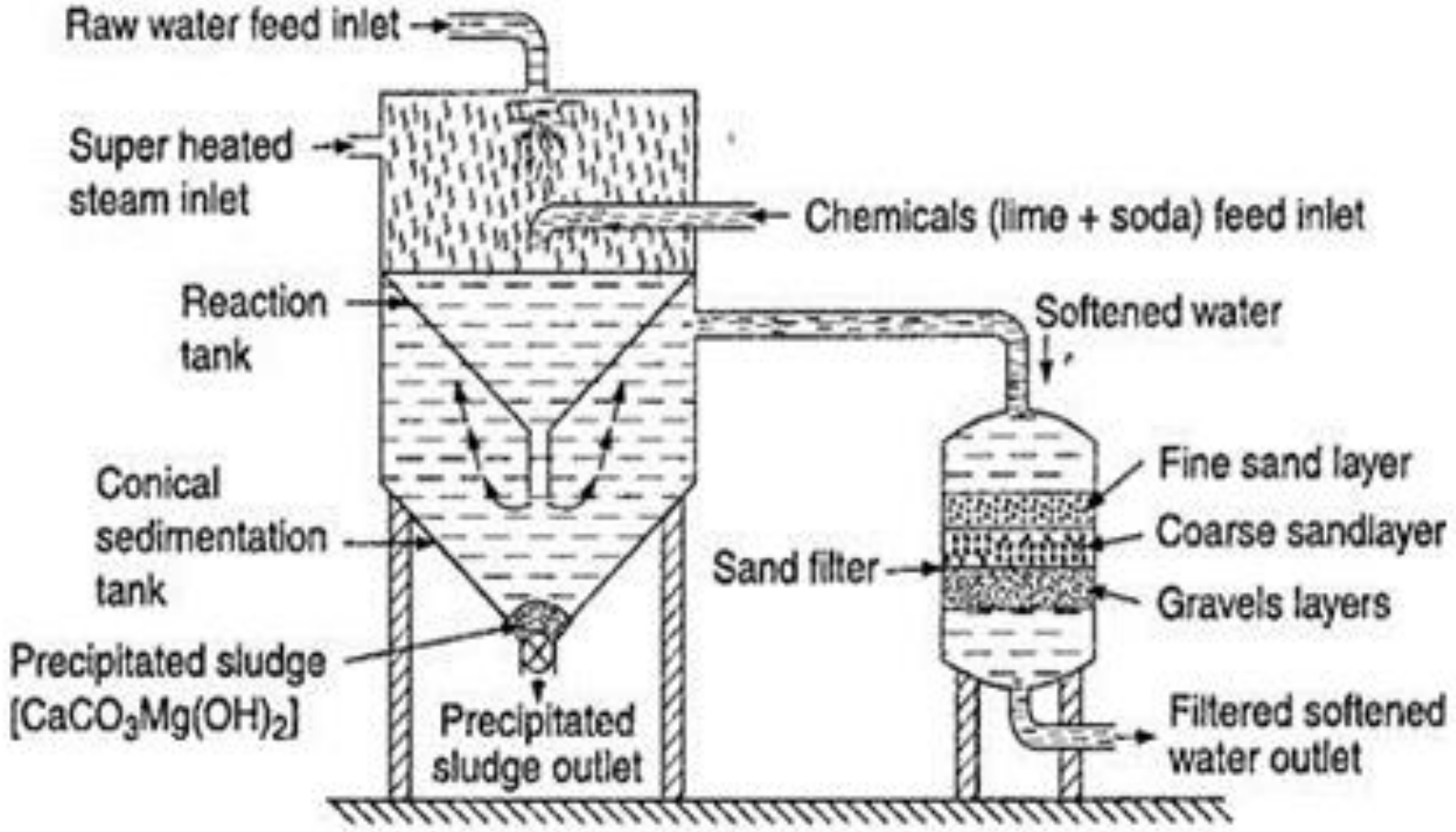
(b) Amount of soda required for softening = 106/100 (Perm. ($Ca^{2+}$ + $Mg^{2+}$ + $Fe^{+2}$ + $Al^{3+}$) + $H^+$ ($HCl/H_2SO_4$) + $HCO_3$ - all in terms of $CaCO_3$ eq.)

# COLD-LIME-SODA PROCESS:



- In this method the lime & soda are mixed with hard water at room temperature with constant stirring.
- Generally the precipitates formed by this process are finely divided and in order to settle the precipitates, coagulants like alum, ferrous sulphate, etc. are added.
- The hard water to be softened is mixed with calculated quantity of chemicals (Lime + Soda + Coagulant) from the top into the inner chamber on vigorous stirring. The chemical reactions takes place and the hardness producing salts get converted into insoluble precipitates.
- The sludge is removed from the bottom of the outer chamber while the softened water passes through a wood fibre filter to ensure the complete removal of any residual sludge particles.
- The clear softened water is withdrawn from the top of the outer chamber.
- The softened water from this process contains a residual hardness of 50-60ppm

# HOT-LIME-SODA PROCESS:



- This process is similar to the cold lime-soda process, but no coagulant is needed.
- Here the process is carried at a temperature of 80$^{o}$ to 150$^{o}$C. Since the reaction carried out at high temperature.

  (a) The reaction takes place faster

  (b) The sludge settles rapidly

  (c) Viscosity of soft water is lower, hence filtered easily

  (d) The dissolved gases such as $CO_2$, air, etc. driven out of the water

  (e) The residual hardness is low, compared to cold lime- soda process.

Hot lime soda process consists of three parts-

- "REACTION TANK" in which complete mixing of water, chemicals and steam takes place and water gets softened.
- "Conical Sedimentation Vessel" where the sludge settle down.
- "SAND FILTER" where sludge is completely removed.

The softened water from this process contains a residual hardness of 15-30 ppm.

## ADVANTAGES OF LIME-SODA PROCESS:

I. This process is economical.
II. Mineral content of the water is reduced.
III. The process increases the pH value of water, which reduces the content of pathogenic bacteria.
IV. Manganese and Iron salts are also removed by this process.
V. The process improves the corrosion resistance of the water.

## DISADVANTAGES OF LIME-SODA PROCESS:

1) Due to residual hardness, water is not useful for high pressure boilers.
2) Large amount of sludge is formed which create disposal problem.

Equivalence

$CaCO_3 = Ca(OH)_2 = Na_2CO_3$

100 = 74 = 106

Lime required = 74/100 [ $Mg^{2+}$ perm. + $HCO_3^-$ + $Ca^{2+}$ temp. + 2 $Mg^{2+}$ temp. + $CO_2$ + $H^+$ + $Fe^{2+}$ + $Al^{3+}$ -$NaAlO_2$] mg/L

$$= \frac{74 [ Mg^{2+} \text{ perm.} + HCO_3^- + Ca^{2+} \text{ temp.} + 2\, Mg^{2+} \text{ temp.} + CO_2 + H^+ + Fe^{2+} + Al^{3+} - NaAlO_2] \times \text{Volume of water} \times 100}{100 \times 1000 \times \% \text{ Purity}} = \text{-------- g of Lime}$$

Soda required

$$= \frac{106 [ Ca^{2+} \text{ perm.} + Mg^{2+} \text{ perm} - HCO_3^- + H^+ + Fe^{2+} + Al^{3+}] \times \text{volume of water} \times 100}{100 \times 1000 \times \% \text{ purity}} = \text{___g of Soda}$$

# Non-hardness causing substances which doesn’t consumes lime – soda:

- KCl
- NaCl
- SiO2
- Na2So4
- Fe2O3
- K2SO4

∴ Lime requirement = ...

Soda requirement = 318 g

(ix) Calculate the quantity of lime and soda required to soften 20,000 liters of water containing following salts:

$CaCO_3$ = 10.0 mg/L  $MgCO_3$ = 8.4 mg/L

$CaCl_2$ = 11.1 mg/L  $MgSO_4$ = 6.0 mg/L

$SiO_2$ = 1.2 mg/L

assuming the purity of lime as 90% and soda as 95%.

**Solution**

**Conversion to $CaCO_3$ equivalent**

| Constituent | Amount of substance (W) | Equivalent wt. of substance (E) | $CaCO_3$ equivalent = $\frac{W \times 50}{E}$ |
|---|---|---|---|
| $CaCO_3$ | 10 mg/L | 50 | 10 mg/L |
| $MgCO_3$ | 8.4 mg/L | 42 | 10 mg/L |
| $CaCl_2$ | 11.1 mg/L | 55.5 | 10 mg/L |
| $MgSO_4$ | 6.0 mg/L | 60 | 5 mg/L |
| $SiO_2$ | 1.2 mg/L | | Does not impart hardness |

$$\text{Lime requirement} = \frac{74}{100}(\text{Temp. } Ca^{2+} + 2 \times \text{Temp. } Mg^{2+} + \text{Perm. } Mg^{2+}) \times \text{Volume of water} \times \text{purity factor}$$

$$= \frac{74}{100}[10+2\times10+5]\times20,000\times\frac{100}{90}$$

Lime requirement = 0.5755 kg

$$\text{Soda requirement} = \frac{106}{100}[\text{Perm.}(Ca^{2+} + Mg^{2+})] \times \text{Volume of water} \times \text{purity factor}$$

$$= \frac{106}{100}[10+5]\times20,000\times\frac{100}{95}$$

$$= 0.3347 \text{ kg}$$

... = 24 mg/L, CaS

**Numerical 1: Calculate the amount of lime (74% pure) & Soda (90% Pure) for softening of 6000 litres of water containing following impurities.**

**$Mg(HCO_3)_2$ =73mg/L, $CaCl_2$ = 222mg/L, $MgSO_4$ = 120mg/L, $Ca(NO_3)_2$ = 164mg/L.**

Soln

| Impurities | Multiplication Factor | $CaCO_3$ eq. | Requirement |
|---|---|---|---|
| 1) $Mg(HCO_3)_2$ = 73mg/L | 100/ 146 or 50/73 | = 73 x 50/73 =50 | 2L |
| 2) $CaCl_2$ = 222mg/L | 100/111 or 50/55.5 | 222 x 100/111 = 200 | S |
| 3) $MgSO_4$ = 120mg/L | 100/120 or 50/60 | 120 x 100/120 = 100 | L + S |
| 4) $Ca(NO_3)_2$ = 164mg/L | 100/164 or 50/82 | 164x 100/164 = 100 | S |

$$\text{Lime required} = \frac{74 \times [\text{CaCo3 eq of 2x } Mg(HCO_3)_2 + MgSO_4] \times \text{volume of water} \times 100}{100 \times 1000 \times \%\ \text{purity}}$$

$$= \frac{74 \times [\ 2 \times 50 + 100] \times 6000 \times 100}{100 \times 1000 \times 74} = 200 \times 6 = 1200g = 1.2Kg$$

$$\text{Soda Required} = \frac{106 \times [\ CaCO_3 \text{ eq of } CaCl_2 + MgSO_4 + Ca(NO_3)_2\ ] \times \text{vol. of water} \times 100}{100 \times 1000 \times \%\ \text{purity}}$$

$$= \frac{106 \times [\ 200+100+100] \times 6000 \times 100}{100 \times 1000 \times 90} = 2826.67g = 2.827Kg$$

Numerical 2: Calculate the quantity of soda and lime needed to soften 10000 litres of water containing following impurities. $Na_2SO_4$ =7mg/L, $SiO_2$ = 2mg/L, $CaCl_2$ = 22.2mg/L, $MgSO_4$ = 2.0mg/L, $Mg(HCO_3)_2$ = 5.0mg/L.

Soln:

| Impurities | Multiplication Factor | $CaCO_3$ eq. | Requirement |
|---|---|---|---|
| 1) $Na_2SO_4$ = 7mg/L | Non-hardness causing | | |
| 2) $SiO_2$= 2mg/L | Non-hardness causing | | |
| 3) $CaCl_2$ = 22.2mg/L | 100/111 or 50/55.5 | = 22.2 x 100/111 =20 | **S** |
| 4) $MgSO_4$ =2.0mg/L | 100/120 or 50/60 | = 2 x 100/120 =1.67 | **L + S** |
| 5) $Mg(HCO_3)_2$ = 5.0mg/L | 100/146 or 50/73 | = 5.0 x 100/146 = 3.42 | **2L** |

$$\text{Lime Required} = \frac{74 \times [\text{CaCo3 eq of } MgSO_4 + 2\text{x } Mg(HCO_3)_2 \times 10000}{100 \times 1000}$$

$$= \frac{74 \times [1.67 + 3.42\text{x}2] \times 10000}{100 \times 1000} = 62.974\text{g}$$

$$\text{Soda Required} = \frac{106 \times [\text{CaCo3 eq. of } CaCl_2 + MgSO_4] \times \text{vol of water}}{100 \times 1000}$$

$$= 229.702\text{g}$$

Numerical 2: Calculate the quantity of soda and lime needed to soften 10000 litres of water containing following impurities. $Na_2SO_4$ =7mg/L, $SiO_2$ = 2mg/L, $CaCl_2$ = 22.2mg/L, $MgSO_4$ = 2.0mg/L, $Mg(HCO_3)_2$ = 5.0mg/L.

Numerical 3: Calculate amount of lime (90% pure) and Soda (98% Pure)for the treatment of 1 million litres of water containing $Ca(HCO_3)_2$ = 8.1ppm, $CaCl_2$=33.3ppm, $HCO_3^-$= 91.5ppm, $MgCl_2$ = 38ppm, $Mg(HCO_3)_2$ = 14.6ppm. The coagulant $Al_2(SO_4)_3$ was added at the rate of 17.1ppm.

| Impurities | Multiplication Factor | $CaCO_3$ eq. | Requirement |
|---|---|---|---|
| 1) $Ca(HCO_3)_2$ = 8.1mg/L | = 100/162 or 50/81 | 5 | L |
| 2) $CaCl_2$ = 33.3mg/L | 100/111 or 50/55.5 | 30 | S |
| 3) $HCO_3^-$ = 91.5mg/L | =100/122 or 50/61 | 75 | **L - S** |
| 4) $MgCl_2$ =38.0mg/L | 100/95 or 50/47.5 | 40 | **L + S** |
| 5) $Mg(HCO_3)_2$ = 14.6mg/L | 100/146 or 50/73 | 10 | **2L** |
| 6) $Al_2(SO_4)_3$ = 17.1mg/L | 100/114 or 50/57 | 15 | **L + S** |

Lime Required = 127444.44g = 127.44Kg

Soda required = 10816.33g = 10.816Kg

# Types of Lime Soda Processes

- Hot Lime-Soda Method
- Cold Lime-Soda Method

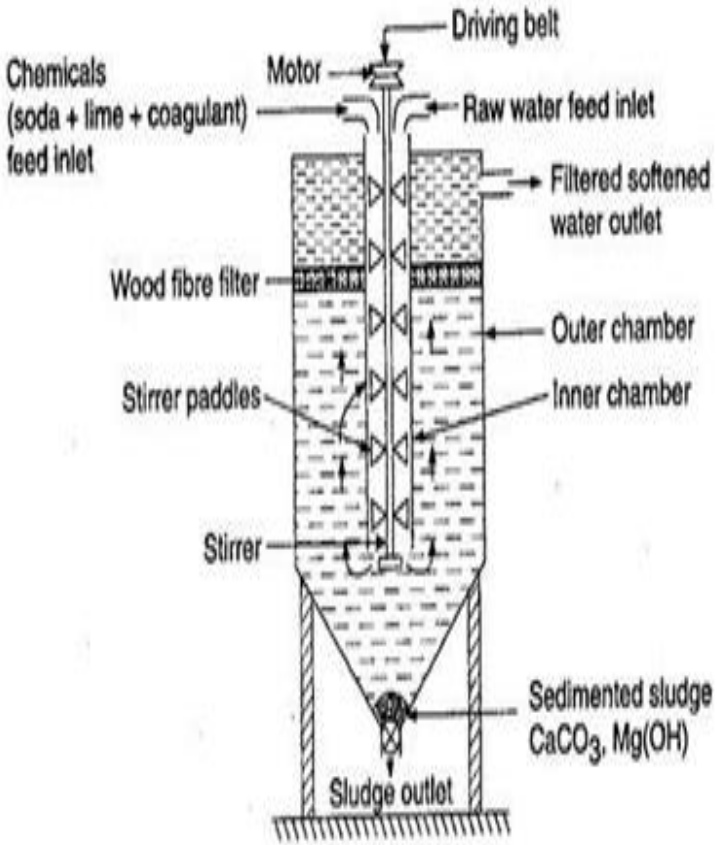
# ➢ Cold Lime Soda Method

## Principle:

1. In this process calculated quantities of lime and soda are mixed
with hard water at room temperature.
2. The **precipitate** formed are finely divided and there fore do not settle down easily and cant be filtered easily.
3. Thus it is essential to add small quantities of coagulants like
alum, aluminum sulphate, sodium aluminates.

These hydrolyze to **gelatinous ppt of aluminum hydroxide** and
entrap fine precipitate.

If sodium aluminate is used as coagulant, silica as well as oil
present in water are also removed.




**Cold lime soda process provides water containing a Residual hardness of 50 to 60 ppm.**

$NaAlO2 + 2H2O \rightarrow NaOH + Al(OH)3$

# ➢ Hot Lime Soda Method

**Principle:**

1.In this method water is treated with softening chemicals like soda and Lime at a **temp. of 80°C to 150°C**. this process is operated at a temp close to boiling point of the solution.

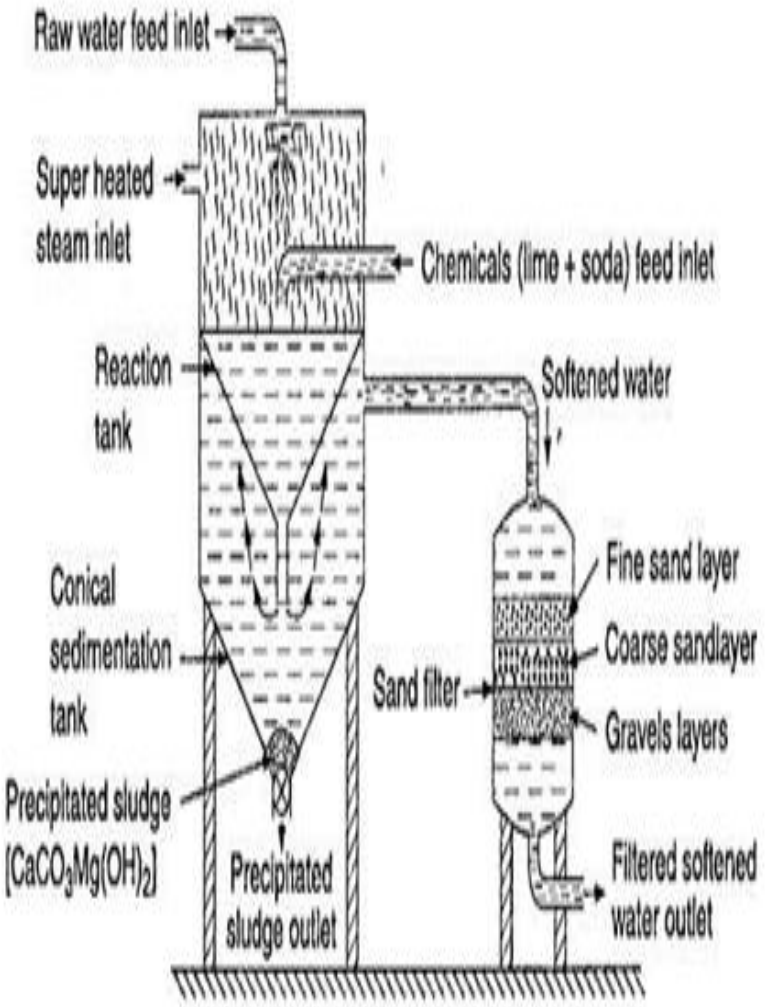
Hence the reaction proceeds faster, the softening capacity.

2. Of the process increased many fold, the precipitate and sludge formed settle down rapidly and hence no coagulants are required.

Much of the dissolved gases like CO2 & air are driven out of water.

Viscosity of soften water is lowered therefore filtration of water becomes much easier.

**This method produces water of comparatively lower residual hardness of 15 to 30 ppm.**



# Advantages of Lime Soda Process

- It is very Economical
- If the process is combined with sedimentation & coagulation, lesser amounts of coagulants shall be needed.
- The process increases the pH value of the treated water, thereby corrosion of the distribution pipe is reduced.
- Besides the removal of hardness, the quantity of minerals in the water is reduced.
- To a certain extent, iron and manganese are also removed from the water.
- Due to alkaline nature of treated water, the amount of pathogenic bacterial in water is considerably reduced.

# Disadvantages

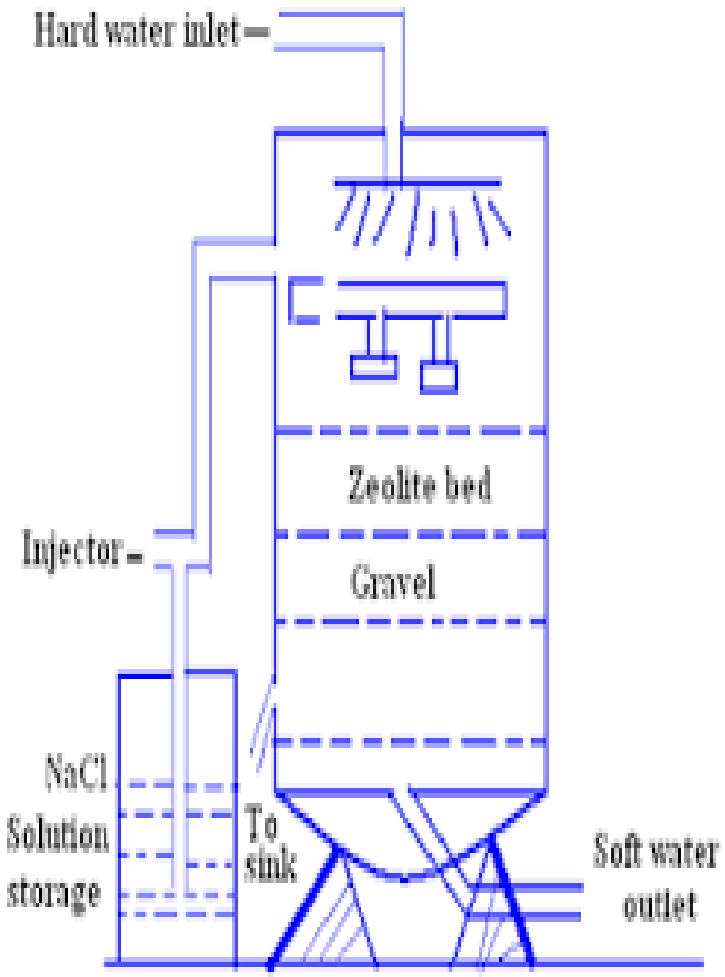
- For efficient and economical softening, careful operation & skilled supervision is required.
- Disposal of large amounts of sludge poses a problem. However the sludge may be disposed off in raising low lying areas of the city.
- This can remove hardness only up to 15ppm which is not good for boilers.

# Softening of water by zeolite process

- Sodium zeolite: $Na_2O.\ Al_2O_3.xSiO_2.yH_2O$
- Where x = 2-10 and y = 2-6
- Zeolite is sodium alumino silicate capable of exchanging reversibly its sodium ions for hardness producing ions present in water.

❑ Types of zeolite

- Natural Zeolite: are non porous eg. Natrolite $Na_2O.Al_2O_3.4SiO_2.2H_2O$
- Synthetic Zeolite: are porous and possess gel structure. They are prepared by heating together china clay, feldspar and soda ash.




Softening Process :

$$Na_2ze + Ca(HCO_3)_2 \longrightarrow CaZe + 2\ NaHCO_3$$

$$Na_2ze + Mg(HCO_3)_2 \longrightarrow MgZe + 2\ NaHCO_3$$

Regeneration :

$$CaZe + 2\ NaCl \longrightarrow Na_2ze + CaCl_2$$

$$MgZe + 2\ NaCl \longrightarrow Na_2ze + MgCl_2$$

## Advantages of zeolite process

- It removes the hardness almost completely and water of about 10ppm hardness is produced.
- The equipment used is compact, occupying small space.
- No impurities are precipitated, so there is no danger of sludge formation in treated water at a later stage.
- The process automatically adjusts itself for variation in hardness of incoming water.
- It is quite clean process.
- It requires less time for softening.
- It requires less skill for maintenance as well as operation.

# Disadvantages of zeolite process

- The treated water contains more sodium salts than in lime soda process.
- The method only replaces Ca2+ and Mg2+ ions by Na+ ions but leaves all the acidic ions like $HCO_3$- and $CO_3$2- as such in the softened water.
- When such soften water is used in boilers for steam generation, sodium carbonate decomposes producing CO2, which cause corrosion and sodium carbonate hydrolyses to sodium hydroxide which cause caustic embrittlement.
- High turbidity water cannot be treated efficiently by this method.

Numerical 1: The hardness of 10000 litres of hard water sample was completely removed by passing it through a zeolite softener. The zeolite softener required 5000litres of NaCl soln containing 1170mg of NaCl/L. Determine hardness of water sample.

Solution:

5000 L of NaCl soln = 5000 x 1170mg of NaCl

= 5850000mg of NaCl

1eq of NaCl = 1eq of $CaCO_3$

58.5mg of NaCl = 50mg of $CaCO_3$

5850000mg of NaCl = $\frac{5000000}{}$ mg of $CaCO_3$

There fore,

10000L of Hard water = 5000000mg of $CaCO_3$

1L of Hard water = 5000000/10000 = 500mg of $CaCO_3$

Hence Hardness of water = 500ppm.

Numerical 2: An Exhausted zeolite softener was regenerated by passing 75L of NaCl soln having strength of 75g/L. If $1.6 \times 10^4$ hard water was passed though zeolite bed, what will be the hardness of water.

Soln:

75L of NaCl = 75 x 75 g of NaCl = 5625g of NaCl

$= 5625 \times 10^3$ mg of NaCl

1eq of NaCl = 1eq. Of $CaCO_3$

58.5mg of NaCl = 50mg of $CaCO_3$

$5625 \times 10^3$ mg of NaCl = $5625 \times 10^3 \times 50/58.5$ mg of $CaCO_3$

$= 4.81 \times 10^6$ mg of $CaCO_3$

$1.6 \times 10^4$ L of hard water = $4.81 \times 10^6$ mg of $CaCO_3$

1L of hard water = $4.81 \times 10^6 / 1.6 \times 10^4$

= 300ppm

Numerical 3: An exhausted zeolite softener was regenerated by passing 125L of NaCl having strength 50gNaCl/L. If the hardness of water is 350ppm, then calculate the total volume of water sample.

Soln:

125L of NaCl = 125 x 50g of NaCl

= 6250g = 6250000mg of NaCl

58.5mg of NaCl = 50mg of $CaCO_3$

6250000mg of NaCl = 6250000x50/58.5

= 5341880mg of $CaCO_3$

1L of hard water = 350mg of $CaCO_3$

Therefore,

x L of Hard water = 5341880mg of $CaCO_3$

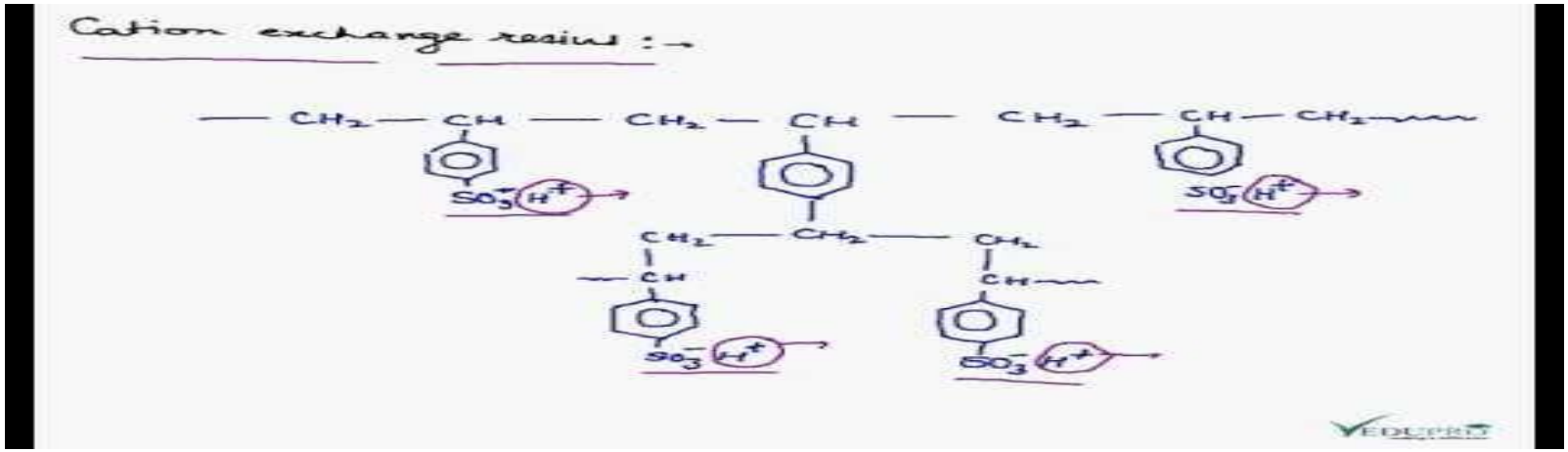
Hence volume of hard water = 5341880/350

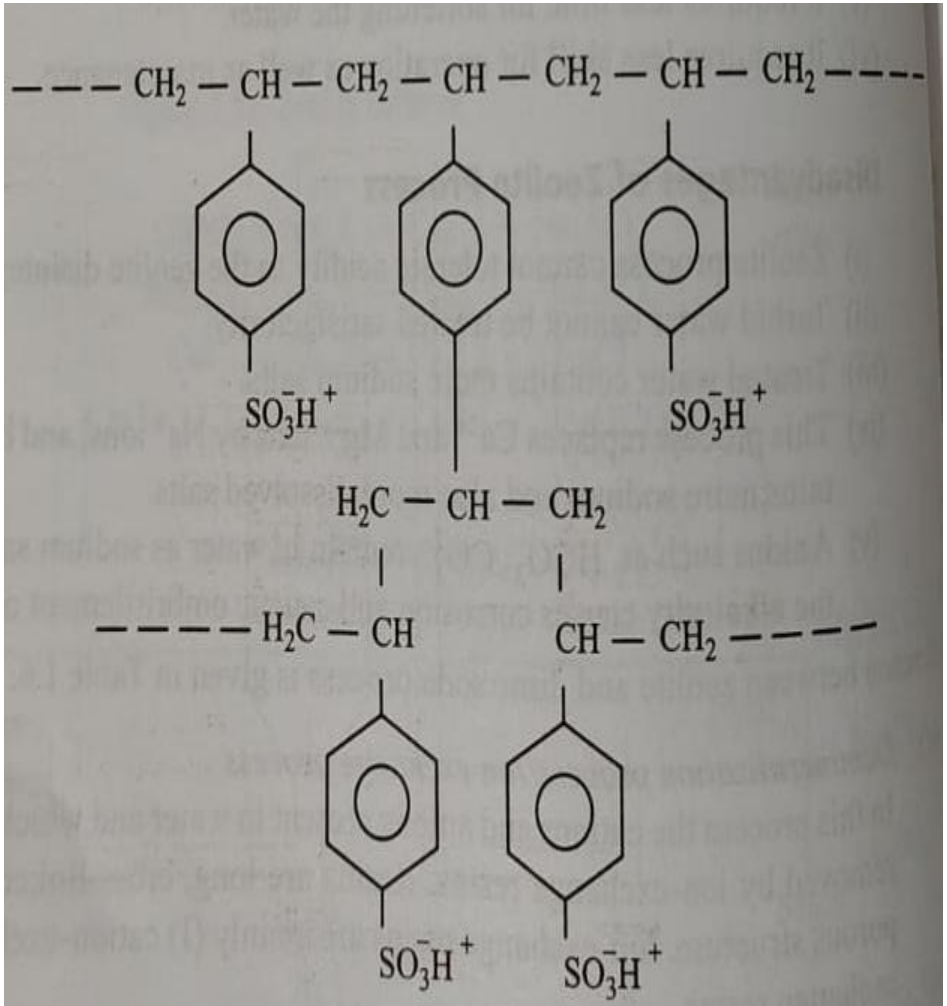
= 15262.5L

- Numerical4: Calculate the volume of NaCl (12.5g/L) in L for regeneration of zeolite, the 200ppm of 10000L of hardwater was passed through it to get zeolite exhausted.

# Softening of Water by Ion-Exchange Process Or Demineralization process

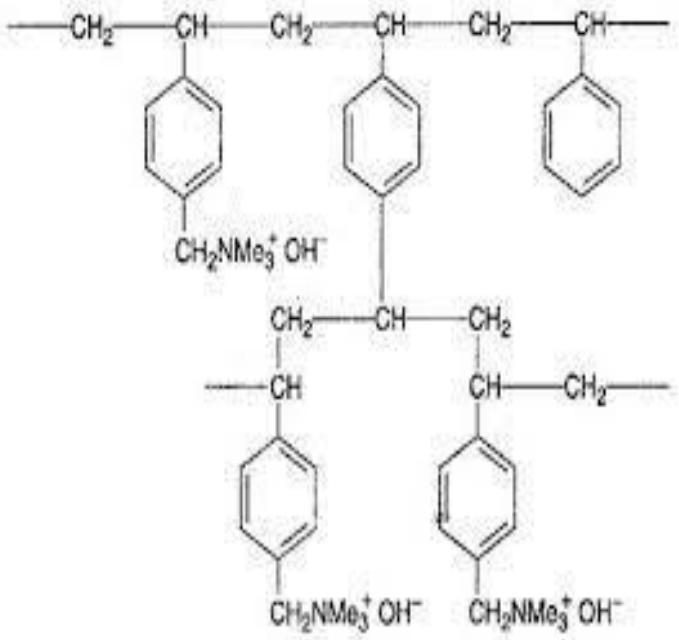
- Ion exchange resins are insoluble cross linked long-chained organic polymers with a microporous structure.
- **Cation Exchange resin:** Styrene-divinyl benzene copolymers which on sulphonation or carboxylation become capable of exchanging their hydrogen ions with the cations in hard water.

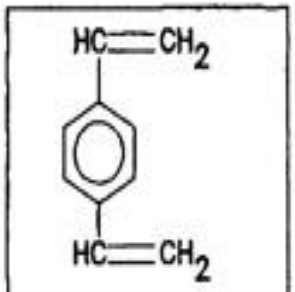




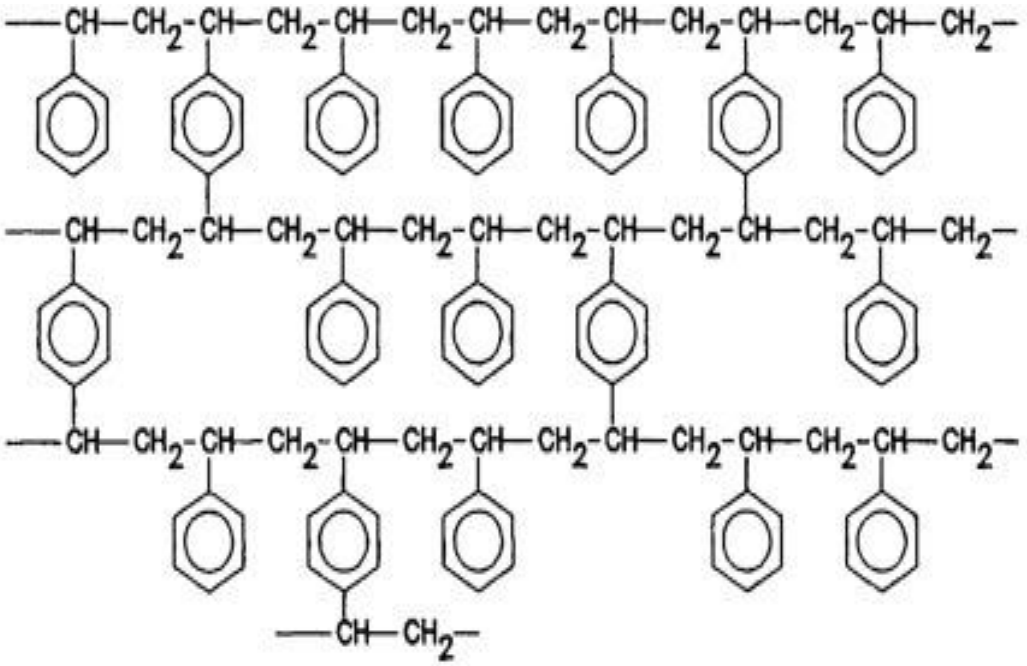

Sulphonation form of cation-exchange resin

**Anion Exchange resin:** Styrene-divinyl benzene copolymers which on further functionalization like formylation followed by Mannich reaction or nitration followed by amination and alkylation gives quaternary ammonium groups as an integral part of the resin matrix. These after treatment with dil NaOH soln become capable of exchanging their OH- anions with anions of hard water.

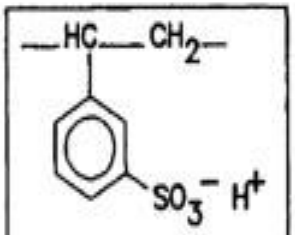




DIVINYL–
BENZENE
MONOMER
FOR CROSS–
LINKING

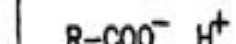


POLYSTYRENE DIVINYLBENZENE BACKBONE



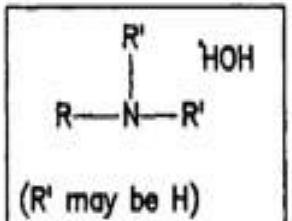
STRONG CATION (SC)

SULFONIC
STRONG ACID
CATION
EXCHANGE
RESIN



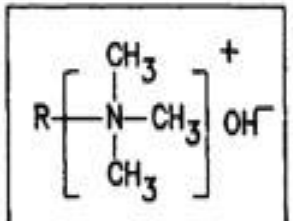
WEAK CATION (WC)

CARBOXYLIC
WEAK ACID
CATION
EXCHANGE
RESIN



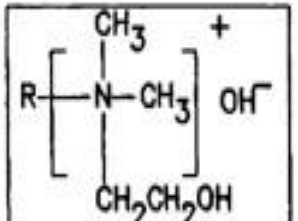
WEAK ANION (WB)

AMMONIUM
WEAK BASE
ANION
EXCHANGE
RESIN



TYPE I STRONG ANION (SB)

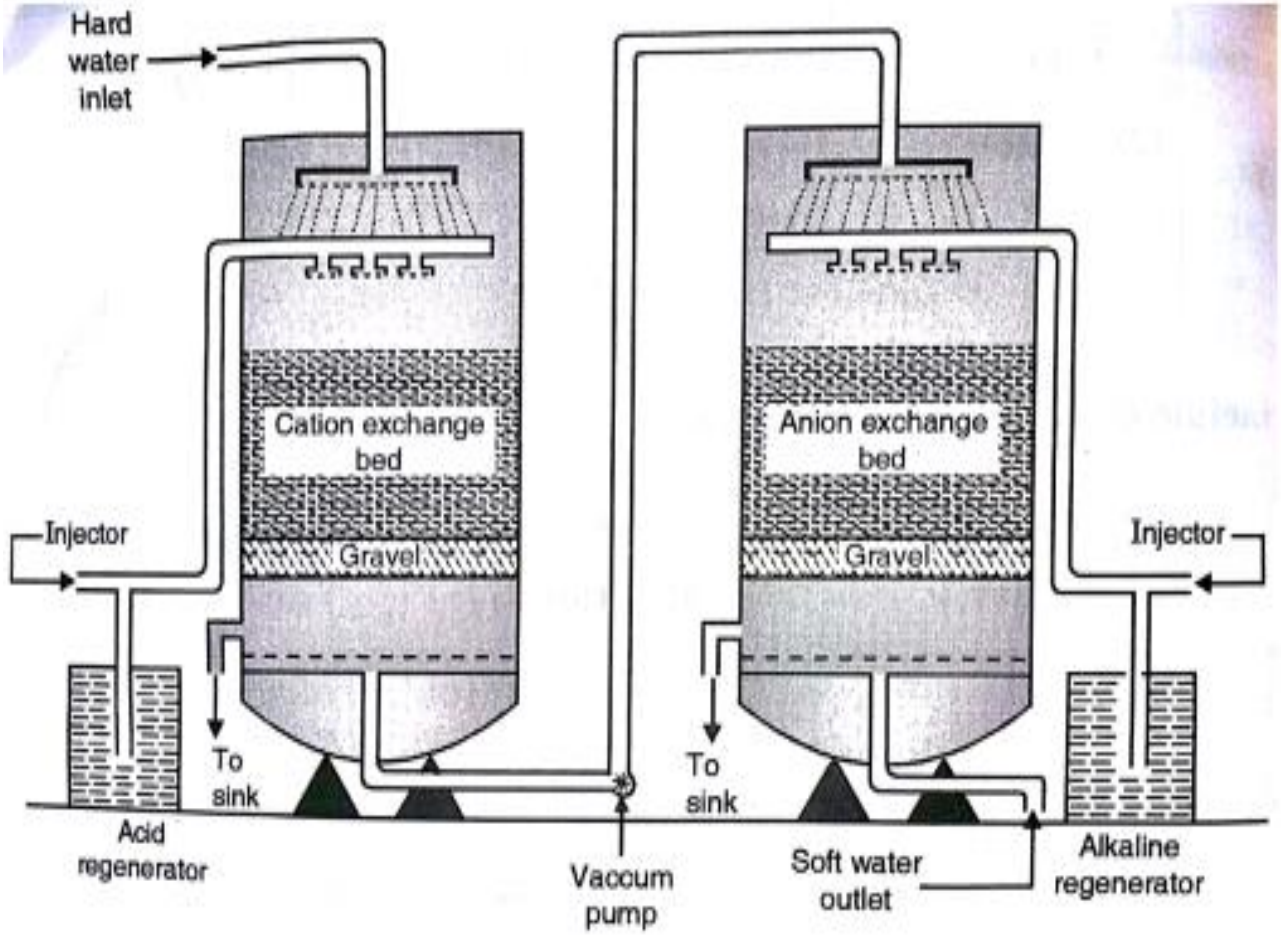
QUATERNARY
AMMONIUM
STRONG BASE
ANION EXCHANGE
RESIN



TYPE II STRONG ANION (SB)

QUATERNARY
AMMONIUM
STRONG BASE
ANION EXCHANGE
RESIN

Process: $Ca(HCO_3)_2$, $CaCl_2$, $MgCl_2$




Demineralisation Diag. 2

**Reactions with Cation Exchanger: RH+**

$RH^+ + Ca^{2+} \rightarrow R_2Ca + 2H^+$

Resin Hardwater Resin Soften water

$RH^+ + Mg^{2+} \rightarrow R2Mg + 2H^+$

$HCO_3^-$, $Cl^-$, $SO_4^-$

**Reactions with Anion Exchanger: R'OH**

$R'OH + Cl^- \rightarrow R'Cl + OH^-$

Resin Hard Water Resin Soften water

$2R'OH + SO_4^- \rightarrow R'2SO4 + 2OH^-$

**Overall Reaction:**

$H^+ + OH^- \rightarrow H2O$

**Regeneration Reactions**

**Cation Exchanger**

$R_2Ca + 2HCl \rightarrow RH^+ + CaCl2$

Regenerated Resin Washings

**Anion Exchanger**

$R'_2SO_4 + 2NaOH \rightarrow 2\ R'OH + Na_2SO_4$

## ➢ Advantages

- The process can be used to soften highly acidic or alkaline water.
- It produces water of very low residual hardness (2ppm) so it is very good for treating water for use in high pressure boilers.

## ➢ Disadvantages:

- The Equipment is costly and more expensive chemicals are needed.
- If water contains turbidity then the output of the process is reduced. The turbidity must be below 10ppm. If it is more it has to be removed first by coagulation and filtration.

Numerical 1: After treating 10000 litres of water by ion exchanger the cationic resin required 200 litres of 0.1N HCl and anionic resin required 200 Litres of 0.1N NaOH soln. Find the hardness of water.

Soln:

Hardness of 10000L of water = 200 L of 0.1N HCl

= 200L of 0.1 N NaOH

1 eq. of HCl = 1 Eq. of $CaCO_3$

0.1N of HCl = 0.1N of $CaCO_3$

Hence,

Hardness of 10000L of water = 200L of 0.1N $CaCO_3$ eq.

= 20L of 1N $CaCO_3$ eq.

Std Statement: 1L of 1N $CaCO_3$ = Eq. wt = 50g

Hardness of 10000L of water = 20 x 1L of 1N $CaCO_3$

= 20 x 50 g = 1000g

= 1000000mg of $CaCO_3$ eq.

Hence,

Hardness of 1L of water = 1000000/10000 = 100mg/L = 100ppm

- Numerical 2: After treating 50000 litres of water by ion exchanger the cationic resin required 150 litres of 0.25N HCl and anionic resin required 150 Litres of 0.25N NaOH soln. Find the hardness of water.

ENVIRONMENTAL ENGINEERING & WATER TESTING

# Water Quality Analysis

Fundamental metrics and laboratory protocols: Dissolved Oxygen, Winkler Titration, BOD, and COD.

CORE WATER QUALITY INDICATOR

# What is Dissolved Oxygen (DO)?

**4–5** mg/L

Minimum dissolved oxygen concentration required for healthy aquatic fish survival.

### Atmospheric Diffusion

Oxygen dissolves naturally from ambient air via turbulence and surface agitation.

### Aquatic Photosynthesis

Submerged plants and algae release gaseous oxygen directly into the water body.

PHYSICOCHEMICAL FACTORS

# Key Factors Controlling Oxygen Solubility

### Water Temperature

Cold water holds higher oxygen levels. Higher temperatures lower solubility.

### Salinity Concentration

Freshwater retains significantly more dissolved oxygen than marine or brackish water.

### Organic Load

Excess organic waste accelerates bacterial growth and rapidly consumes oxygen.

CHEMICAL ANALYSIS TECHNIQUE

# The Winkler Method: Overview

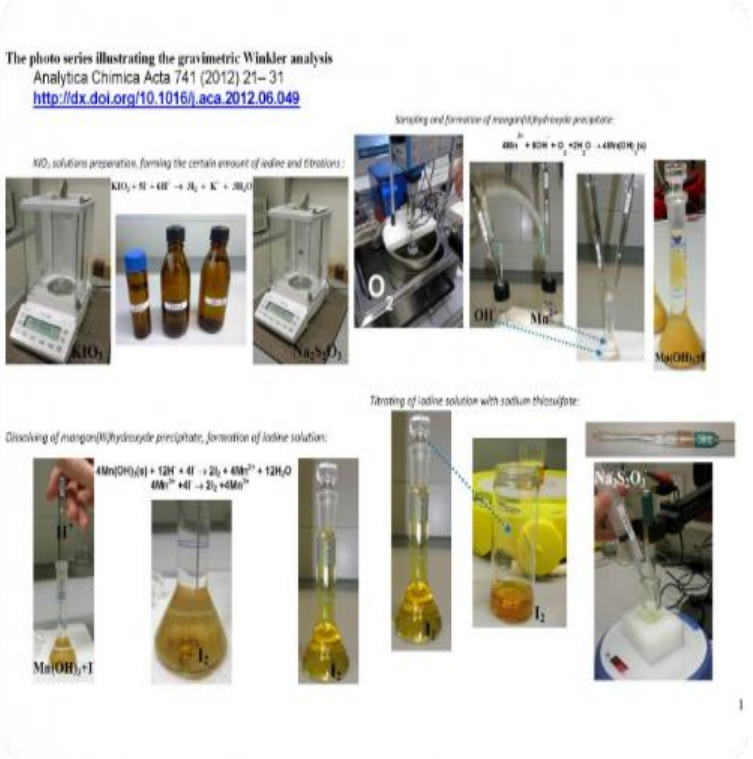
A classic iodometric titration procedure used to determine precise concentration of dissolved oxygen in water.

### Fixing Strategy

Dissolved oxygen is chemically bound into stable manganese compounds to prevent gaseous escape.

### Stoichiometric Stopper Ratio

1 mole of dissolved $O_2$ yields 2 moles of free iodine during titration.



LABORATORY PROCEDURE

# Winkler Titration: 4-Step Process

1

## Sample Fixing

Add $MnSO_4$ and alkali-iodide reagent. Brown precipitate forms.

2

## Acidification

Add concentrated $H_2SO_4$. Floc dissolves, releasing free iodine.

3

## Thiosulfate Titration

Titrate with $Na_2S_2O_3$ until golden yellow turns pale straw.

4

## Starch Endpoint

Add starch indicator. Titrate blue solution to clear endpoint.

ORGANIC POLLUTION METRIC

# Biochemical Oxygen Demand ($BOD_5$)

### $BOD_5$ Calculation Formula

$$BOD_5\ (mg/L) = (DO_b - DO_i) \times \text{Dilution Factor}$$

**DO_b (Initial DO)**

Dissolved oxygen measured immediately after sample dilution.

**DO_i (Incubated DO)**

Dissolved oxygen measured after 5 days incubation at 20°C.

RAPID TOTAL OXIDATION

# Chemical Oxygen Demand (COD)

### COD Calculation Formula

$$COD\ (mg/L) = [(V_1 - V_2) \times N_FAS \times 8000] / Sample\ Volume\ (mL)$$

**$V_1$ (Blank)**

FAS volume titrated for distilled water blank.

**$V_2$ (Sample)**

FAS volume titrated for wastewater sample.

**8000 Factor**

Oxygen equivalent mass constant (8 g/eq × 1000).

COMPARATIVE ANALYSIS

# BOD vs. COD Comparison

| Feature | Biochemical Oxygen Demand (BOD) | Chemical Oxygen Demand (COD) |
|---|---|---|
| **Oxidation Agent** | Aerobic Microorganisms | Potassium Dichromate ($K_2Cr_2O_7$) |
| **Test Duration** | 5 Days (at 20°C) | 2 to 3 Hours |
| **Organic Scope** | Biodegradable Organics Only | Total Organics (Bio + Non-biodegradable) |
| **Value Relation** | Lower Value (BOD < COD) | Higher Value (COD > BOD) |

PRACTICAL APPLICATION

# Applications in Wastewater Engineering




### Plant Efficiency Monitoring

BOD removal rates indicate treatment plant efficiency.

### Industrial Discharge Compliance

COD tests provide rapid compliance checks for toxic effluents.

### Stream Self-Purification

DO sag curves model natural oxygen recovery in rivers.

## ➢ BOD:

Biological oxygen demand of sewage is defined as the amount of free oxygen required for the biological oxidation of the organic matter under aerobic condition at 20°C for a period of 5days.

Unit of BOD = ppm or mg/L

- Determination of BOD

1. A known volume of effluent sample is diluted with a known volume of water. Sample is diluted to ensure that sufficient oxygen is available for the complete aerobic oxidation of organic matter.
2. The diluted sample is divided into two parts and kept in two stoppered bottles. The dissolved oxygen content of one of the bottle is determined by Winkler's method, which is blank reading. It is denoted as $DO_b$ .
3. Another bottle is incubated at 20°C for 5days. Then dissolved oxygen is determined. This is denoted as DOi.

The difference between DOb and DOi gives BOD.

BOD = (DOb-DOi) x dilution factor

= (DOb-DOi) x Volume of sample after dilution/volume of sample before dilution.

BOD test is usually influenced by the type of micro-organisms, presence of toxins, pH, Some reduced minerals; etc.

Reaction:

Organic Matter + $O_2$ → $CO_2$ + $H_2O$

BOD is proportional to the amount of organic waste to be degraded.

An average sewage has BOD of **100-150** ppm.

Drinking water should have BOD less than 1ppm.

Significance of BOD:

1. It indicates the amount of decomposable organic matter in sewage.
2. The higher is the BOD of a sample the higher the amount of decomposable organic matter in the sample and the higher is the pollution in water.
3. From BOD self purifying capacity of streams cans be determined. Thus it provides the degree of pollution at any time in the sewage stream.

## ➢ COD

Chemical oxygen demand may be defined as the amount of oxygen required for the oxidation of chemically degradable organic matter in the hot conditions with the use of oxidants like acidified $K_2Cr_2O_7$ in 2-3hrs.

Unit of COD = mg/L or ppm

- Determination of COD:

1. A known volume of waste water sample is refluxed with a known excess of standard potassium dichromate soln and dilute sulphuric acid in presence of silver sulphate for 2-3hs. Silver sulfate is used as catalyst to promote oxidation of straight chain aliphatic compounds, aromatic compounds.
2. The organic matter of the sample is oxidized to ammonia, $CO_2$ and $H_2O$. The unreacted $K_2Cr_2O_7$ is titrated against Ferrous ammonium sulphate[$FeSo_4.(NH_4)_2SO_4.H_2O$]i.e. Mohr's Salt.
3. This experimentally measured amount of potassium dichromate is used to calculate the equivalent oxygen required by the waste water for degradation of the pollutants.

Reactions

$C_xH_yO_z + (x + y/4 - z/2)\ O_2 \rightarrow x CO_2 + y/2\ H_2O$

The unreacted dichromate solution is titrated against Mohrs salt using ferroin indicator.

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$[Fe^{2+} \rightarrow Fe^{3+} + e] \times 6$

Overall reaction

$Cr_2O_7^{2-} + 14H^+ + 6\ Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$

The COD is calculated as

$$COD = \frac{(V1-V2) \times N_{FAS} \times 8000}{\text{Volume of sample}}$$

V1 = vol. of FAS in blank titration

V2 = vol. of FAS in back titration

# Significance of COD

The COD value is not affected by the presence of toxins and other unfavaourable conditions for the growth of microorganisms.

It is taken as basis for calculation of efficiency of treatment plant.

It help in designing the water treatment plant.

Numerical 1: A 50mL of sample contains 840ppm of dissolved oxygen. After 5days the dissolved oxygen value becomes 230 ppm, after the dilution of sample to 80mL. Calculate the BOD.

Soln: BOD = (Dob-Doi) x Volume of sample after dilution/ vol of sample before dilution

BOD = (840-230) x 80/50 = 976ppm

Numerical 2: A 25mL of sewage water sample was refluxed with 10mL of 0.25N $K_2Cr_2O_7$ soln in presence of dil.$H_2SO_4$, $Ag_2SO_4$ and $HgSO_4$. The unreacted dichromate required 6.5ml of 0.1N FAS. 10mL of same $K_2Cr_2O_7$ required 27mL of 0.1N FAS. Calculate COD.

Given: Volume of sample = 25mL

Blank titration reading (V1) = 27mL

Back titration reading (V2) = 6.5mL

COD= (V1-V2) x N(FAS) x 8000/Vol. of Sample

= (27-6.5) x 0.1 x 8000/25 = 656ppm

Numerical 3: A 30mL of sewage water sample was refluxed with 15mL of 0.25N $K_2Cr_2O_7$ soln in presence of dil.$H_2SO_4$, $Ag_2SO_4$ and $HgSO_4$. The unreacted dichromate required 14 ml of 0.05N FAS. 15mL of same $K_2Cr_2O_7$ required 20mL of 0.1N FAS. Calculate COD.

Note: Covert one statement in to fixed normality.

Volume of sample = 30mL

Blank Titration reading (V1) = 20ml of 0.1N FAS = **40 mL** of **0.05N** FAS

Back Titration reading (V2) = **14mL** of **0.05N** FAS

COD = 346.667ppm

SUMMARY

# Key Takeaways

### DO (State)

Measures currently available oxygen supporting aquatic biology.

### BOD (Biological)

Quantifies biodegradable waste load over 5 days incubation.

### COD (Chemical)

Measures total organic contamination within hours.

# Sewage Treatment

A sewage is the liquid waste which includes human and house hold waste water, industrial waste ground waste and street, storm waste.

Before disposing off the sewage to river or sea, it has to be treated.

To prevent –pollution of water

-- Offensive odour in water

- the destruction of fish and other aquatic life

Sewage treatment by activated sludge process.

Sewage contains minerals and inorganic matter in suspension and in solution. It contains sometimes harmful living organisms. Hence it is better to treat sewage before releasing into the river, lake and sea.

It involves four steps

1. Preliminary treatment
2. Primary treatment’
3. Secondary or biological treatment: activated sludge method
4. Tertiary treatment

1) Preliminary Treatment

The main objective of preliminary treatment is the removal of gross solid i.e. large floating and suspended matter, oil etc. For removing inorganic matter, sewage is allowed to pass through bar screens and mesh screens. It is also called as screening of sewage.

2) Primary Treatment

For removing suspended matter efficiently and economically, sedimentation process is carried out.

The sewage is treated with certain chemicals like(alum, hydrated lime etc.) which form a floc that absorbs and entrains the suspended and colloidal particles present.

3) Secondary treatment or Biological treatment

It is essentially an aerobic oxidation which includes filtration and activated sludge process.

i) Filtration: in this process, sewage water is filtered through specially designed sprinkling filters so that aerobic conditions are maintained.

During this process carbon of the organic matter is converted to $CO_2$ and nitrogen into $NH_3$ and finally onto nitrites and nitrates.

Trickling filters are used for the biological oxidation of sewage. Sewage is delivered to the filters by means of rotating distributors. As the trickled sewage starts flowing downwards, the aerobic bacteria grow on the surface. The bacteria bring about the biological oxidation of organic matter of sewage. The treated water comes out from bottom.

ii) Activated sludge process

The sludge formed from previous oxidation process is known as activated sludge.

This process consists of mixing of sedimented sewage with proper quantity of activated sludge and the mixture is then agitated for 4-6hrs by blowing air.

Suspended dissolved organic matter is oxidized by aerobic oxidation, then affluent is sent to sedimentation tank, where sludge is deposited and clean water drawn off.

A part of  sludge deposited is used for next oxidation batch and the remainder is either spread on land as fertile matter or used for biogas or dumped in sea.

4) Tertiary Treatment

It is applied to decrease the  load of nitrogen and phosphorous compounds present in effluents.

It involves treatment of effluent with lime which reacts with phosphorous to form calcium phosphate.

Nitrogen stripping is carried out to remove ammonia gas.

Chlorination was carried out to kill the disease causing microorganisms present in sewage water.

The so treated water is finally discharged into the rivers or lakes.